

Path Tracking and Handling Stability Control Strategy with Collision Avoidance for the Autonomous Vehicle under Extreme Conditions

Yong Chen, Sizhong Chen, Hongbin Ren, IEEE member, Zepeng Gao, Zheng Liu

Abstract—A novel control strategy is proposed for the autonomous vehicle equipped with the front and rear independent steering systems to enhance the capability of collision avoidance and improve path tracking and handling stability under critical maneuvers. The novel control system includes a path replanning controller, a path tracking controller, and a stability controller. The path replanning controller is designed to generate a collision-free trajectory by optimizing a cost function. The cost function can be updated quickly and iteratively, according to the dynamic fitting matrix and measurement signals. The path tracking controller and the stability controller with considering multi-constraints are designed to generate the steering angle of the front wheel and the rear wheel, respectively, which improve the ability of path tracking and handling stability of the autonomous vehicle under extreme conditions. The real-time effectiveness of the proposed control strategy is verified and evaluated in Hardware-in-the-loop (HIL) simulation environment. The function of collision avoidance of the control algorithm is verified in the scenario of the preset obstacle artificially on the globally planned path. The proposed control strategy, presents superior performance against two other control strategies based on model predictive control algorithm, and also exhibits satisfactory performance in terms of path tracking as well as handling stability of the autonomous vehicle.

Index Terms—Autonomous vehicle, Path tracking, Handling stability, Collision avoidance

I. INTRODUCTION

Owing to the increasing demand for the transportation market, especially the intelligent transportation field, the innovation of internet technology, and the gradual maturity of artificial intelligence technology, autonomous vehicles have witnessed the explosive increase and developed more vigorously in the automobile industrialization over the past few decades. It brings a series of problems, especially energy scarcity and road safety issues [1-3]. The technologies associated with autonomous cars are worth studying in vehicle research institutions and automobile companies. Autonomous vehicles have a broad application prospect in the future [4-6].

Copyright (c) 2015 IEEE. Personal use of this material is permitted. However, permission to use this material for any other purposes must be obtained from the IEEE by sending a request to pubs-permissions@ieee.org. This work was supported in part by the National Nature Science Foundation of China under Grant 51375046, 51205021 and 52002025, and in part by the National Key Research and Development Program of China under Grant 2017YFB0102602. The review of this paper was coordinated by Editors of XXX. (Corresponding author: Hongbin Ren, renhongbin2106@126.com.)

The authors are with the School of Mechanical Engineering, Beijing Institute of Technology, Beijing 100081, China (e-mail: chenyong6011@163.com; chensz@bit.edu.cn; renhongbin2106@126.com; gaozpeng@bit.edu.cn; mailliu Zheng@gmail.com).

The most up-to-date driving systems, from the earlier advanced driver assistance system (ADAS), to adaptive cruise control (ACC) system [7], and to the latest combined active front steer (AFS) and direct yaw control (DYC) system [8-9] have been successfully deployed in the intelligent vehicles. Those associated technologies can comprehensively enhance the enormous potential of vehicle dynamics capacity in safety and handling stability [10]. However, the research of highly automated vehicles is still confronted with considerable challenges from environmental perception [11-13], to path planning [14-16], and to path tracking and vehicle handling stability control [17-18].

The path planning task plays an important role in autonomous vehicle technologies. The goal of the task is to find the optimal path under specific scenarios. It not only needs to generate the high-quality path in real-time, but also needs to weigh the path quality, efficiency, and consumed computing resources of the path comprehensively under the premise of meeting the requirements of vehicle dynamics and environmental constraints [18]. To generate the desired trajectory, Ji *et al.* constructed a 3-D virtual dangerous potential field to avoid the possible collision with obstacles, which rely on a large amount of calculation of derivation of the collision-free trajectory [16].

As the core issue of automated driving technology, obstacle avoidance plays a critical role in vehicle safety [19-20]. To avoid potential collisions and also satisfy vehicle dynamics requirements, the path planning algorithm has to consider about dynamic or static obstacles and vehicle maneuvering capability, comfortability, and also handling stability [21-22]. A collision-free path of the vehicle is planned and updated in real-time based on kinematics and dynamics of the vehicle system when it suddenly encounters emergency scenarios [22]. Li *et al.* treated the optimal collision avoidance path as a problem finding the optimal solution for an objective function considering both safety and comfort performance when assessing the generated trajectories and selecting the optimal reference path [19]. When a series of requirements of vehicle dynamics, constrained environments, and obstacle avoidance are considered in the path planning algorithm, the calculation efficiency is difficult to guarantee, there are potential conflicts among the various objectives. To deal with the conflict between vehicle stabilization actions and the necessary collision avoidance in emergency scenarios, Funke *et al.* presented a new control structure that coordinates those conflicts by designing the priority algorithm [21]. Similarly, if automatic braking is used to avoid vehicle collision accidents in emergency scenarios, it can affect the lateral stability. To coordinate the longitudinal motion and the lateral motion, Cheng *et al.*

Copyright (c) 2015 IEEE. Personal use of this material is permitted. However, permission to use this material for any other purposes must be obtained from the IEEE by sending a request to pubs-permissions@ieee.org.

proposed a lateral-stability-coordinated collision avoidance control system by adopting three different control modes [23].

The path tracking control commands the actuators to follow the planned route, and the desired speed profile is obtained by the trajectory generation [24]. The path tracking algorithms, e.g., PID-based, feedforward and feedback [25-26], model predictive control (MPC) [19][27], robust control [28] and optimal preview method [29-30] have been utilized for automated ground vehicles. Falcone *et al.* presented an MPC approach based on the simplified vehicle model to track the desired path by controlling an active front steering system [31]. For the autonomous vehicles at limits of handling, there is a considerable challenge to satisfy the collision avoidance and stabilization criteria at the same time because the tire has nonlinear characteristics in this case. The G-G diagram and the bicycle model are used in a race car to imitate drivers making steering and throttle corrections by the lane-keeping steering feedback controller [32-33]. Besides, to improve the performance of the feedforward controller, Ni *et al.* proposed an adaptive real-time estimation for the tire cornering stiffness based on the predefined Magic Formula model [33].

Path tracking control of the autonomous vehicle is closed related to vehicle dynamics control, e.g., vehicle safety and handling stability. Therefore, this is the core and critical problem of the chassis integrated control of the autonomous vehicle when path tracking of the autonomous vehicle and vehicle dynamics control are taken into account simultaneously. Hu *et al.* proposed a novel output constraint controller to deal with a path-following problem for the four wheels independently driving (4WD) autonomous vehicle, aiming to maintain the lateral stability by adopting an adaptive and robust linear quadratic regulator [34]. Chen *et al.* also proposed a simultaneous path following and lateral stability control method for 4WD and four-wheel independent steering (4WS) autonomous vehicles. The control demand of path following and vehicle stability is obtained in the upper layer controller based on Hamilton energy function control theory [35]. Ji *et al.* utilized a simplified linear bicycle model for path tracking with constraints of the small tire slip angle [16]. Shengbo *et al.* improved path tracking based on the linear MPC method by compensating for the tire nonlinearity. The effectiveness of the control algorithm is verified by the co-simulation environment based on Simulink and Carsim [36]. However, it is not evaluated in the real-time simulation environment or experimental vehicles.

Traditionally, collision avoidance manoeuvre is considered as complex environmental constraints in the path planning or as constraints of vehicle motion from a driving safety perspective [37]. In fact, except for the limitation of obstacles, the minimal lateral offset of an updated replanning trajectory, and the maximum adhesion margin of the road, should be simultaneously considered in generating a collision-free path. For path following and handling stability control, most previous works have been investigated to improve vehicle safety and minimize lateral offset error at normal working conditions [38]. However, under the limit condition, the lateral acceleration of the vehicle should limit the maximum allowable lateral acceleration confined by vehicle dynamics and adhesion coefficient. Path following in critical situations may pose problems for vehicle handling stability because of the nonlinear

characteristics of the tire force and the influence of the sideslip angle of center gravity (C.G) [39-40]. The path tracking and handling stability should be considered comprehensively in evaluating the overall performance of the highly automated vehicle under critical conditions.

The main original innovations and scientific contributions concentrate on the following several points. Firstly, to avoid obstacles and ensure the handling stability of the vehicle in limit conditions, a new collision-free path is generated in real-time in the path replanning controller by a cost function. It allowed temporarily deviate from desired paths with the lateral acceleration of the vehicle limiting to the maximum allowable lateral acceleration on the low adhesion road. Secondly, the prediction error model, in the path tracking controller, is used to minimize the tracking deviation error by an objective function. The maximum lateral offset in various constraints is set to better adapt to the handling stability of the vehicle under the limit conditions. Thirdly, in the stability controller, the sideslip angle of C.G is controlled to prevent the vehicle sideslipping by coordinating the rear-wheel steering angle. In addition, the tire cornering stiffness is estimated to improve the controller's robustness. The real-time and effectiveness of the vehicle control strategy is verified in the hardware in the loop (HIL) real-time simulation environment platform based on dSPACE/Autobox and vehicle controller.

The remainder of the paper is organized as follows. The basic configuration of the autonomous vehicle and the dynamic modeling used for the proposed control strategy is introduced in Section II. The path tracking and handling stability control strategy with collision avoidance of the autonomous vehicle is detailed in Section III. In Section IV, a series of experimental simulation tests are carried out, which intend to demonstrate the performance in terms of path tracking and handling stability as well as collision avoidance under extreme scenarios. The conclusion, summarized in Section V, is concisely remarked.

II. SYSTEM CONFIGURATION AND CONTROL-USED MODELLING OF AUTONOMOUS VEHICLES

A. The system configuration of the autonomous vehicle

This section mainly introduces the system configuration of the autonomous vehicle and nonlinear vehicle dynamics model applied for control design. In vehicle dynamics modeling, Magic Formula with variable tire-road coefficient adhesion under combined conditions is introduced to determine the tire longitudinal/lateral force.

Fig. 1 shows the system configuration of the autonomous vehicle, which is equipped with intelligent equipment and actuators associated with those technologies from environment perception, to path planning, and to the path following and vehicle dynamics control.

The surrounding environment information and precise localization of the autonomous vehicle are achieved by lidar, video camera, Global Navigation Satellite System and Inertial Navigation System (GNSS/INS) based on multi-sensor fusion technology and the high-precision map. The desired trajectory of the vehicle is planned in the path planning layer in Industrial Personal Computer (IPC).

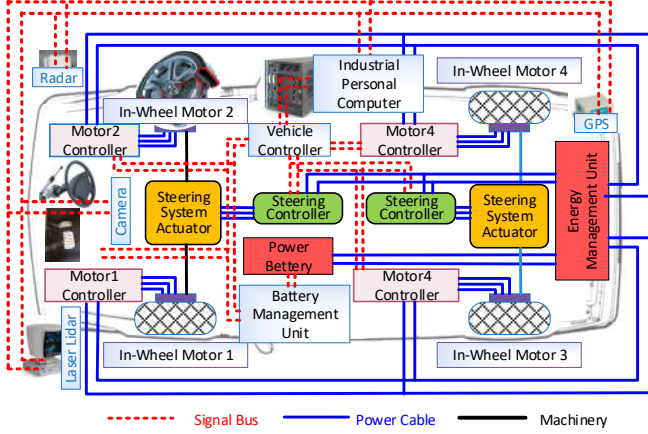


Fig. 1. The system configuration of the autonomous vehicle.

The steering system of the vehicle includes the front wheel steering system and the rear-wheel steering system. It receives control demand signals from the front wheel steering system controller and the rear-wheel steering system controller, respectively. Both controllers receive control signals from the vehicle control unit embedded in the proposed control strategy so that the autonomous vehicle meets the requirements of dynamics and path tracking.

In our previous work [41], the handling stability has been studied for human drive vehicle equipped with four in-wheel motor. The chassis driving system of the car is applied to the autonomous vehicle.

B. Dynamic modeling of the autonomous vehicle

A multiple-degree-of-freedom dynamics model of the autonomous vehicle is built with the tire model varying with the tire-road adhesion coefficient based on Magic Formula. The dynamics model of the autonomous vehicle considering the influence of the road adhesion coefficient is the theoretical basis of the proposed control strategy under extreme conditions, such as the icy and snowy road [41].

Fig. 2 shows the aerial view of the autonomous vehicle in which l_f , l_r , l , and d represent the shortest distance from the front axle to C.G., the shortest distance from C.G. to the rear axle of the vehicle, wheelbase, and half of the track, respectively. u , v , and ψ denote the longitudinal speed, the lateral speed, and the yaw angle of the vehicle in the vehicle coordinate system, respectively. β is the sideslip angle of C.G. F_{xij} and F_{yij} are the longitudinal force and lateral force of each tire in the tire coordinate system where the subscript ij ($i = f$ or r ; $j = l$ or r) indicates which tire of the vehicle. $i = f$ and $i = r$ represent the tire of the front axle and the tire of the rear axle of the car while $j = l$ and $j = r$ denote the left side tire and the right side tire of the vehicle, respectively. The following description of the footnote ij has the same meaning. δ_f , δ_r , and r denote the steering angle of the front wheel, the steering angle of the rear wheel, and the yaw rate of the autonomous vehicle, respectively.

The nonlinear dynamics model of the autonomous vehicle with consideration of tire properties is depicted as

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}) \\ \mathbf{y} = \mathbf{c}\mathbf{x} \end{cases} \quad (1)$$

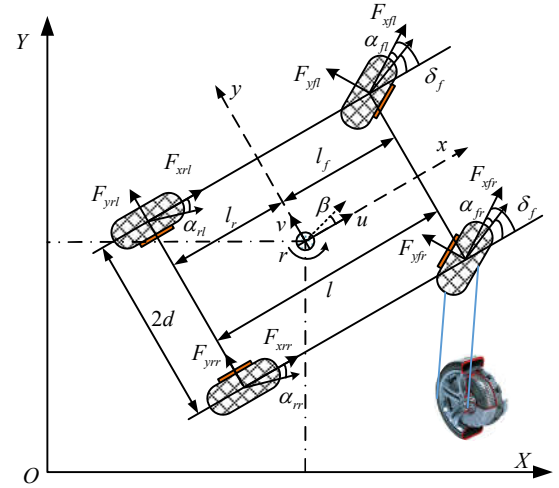


Fig. 2. The aerial view of the autonomous vehicle.

where, the vector $\mathbf{x} = [v \ u \ \psi \ r \ Y \ X]^T$ represents the system state vector. The vector $\mathbf{u} = [\delta_f \ \delta_r]^T$ and the vector $\mathbf{y} = [\psi \ Y]^T$ represent the input vector and the output vector of the vehicle system, respectively. Equation (1) can be described in detail by (2)-(12).

Two coordinate systems XOY and xoy represent the geodetic coordinate system and vehicle coordinate system, respectively. The vehicle centroid position in the absolute coordinate frame (X, Y) is obtained by coordinate frame by

$$\begin{cases} \dot{X} = u \cos \psi - v \sin \psi \\ \dot{Y} = u \sin \psi + v \cos \psi \end{cases} \quad (2)$$

where, X and Y represent the coordinate points of the vehicle's centroid in the absolute coordinate system. The following details the model of the nonlinear dynamics by planer motion and the tire model.

Owing to path tracking and handling stability of the vehicle are closed to planer motion of the car, the differential equations of planer motion, which include longitudinal, lateral and yaw directions, are established [41].

$$\begin{bmatrix} m_v(\dot{u} - vr) \\ m_v(\dot{v} + ur) \\ I_z \dot{r} \end{bmatrix} = \mathbf{Q}_1 \mathbf{F}_x + \mathbf{Q}_2 \mathbf{F}_y + \mathbf{d} \quad (3)$$

where, m_v and I_z are, respectively, vehicle mass and moment inertia of the vehicle. $\mathbf{Q}_1$ and $\mathbf{Q}_2$ are matrices, which are related to the tire force in two directions, respectively. $\mathbf{F}_x$ and $\mathbf{F}_y$ are matrices, which consist of the longitudinal force and the lateral force of the tire, respectively. $\mathbf{d}$ is the external disturbances composed of the wind resistance and the roll resistance. To be more specific, $\mathbf{Q}_1$, $\mathbf{Q}_2$, $\mathbf{F}_x$ and $\mathbf{F}_y$ can be described as (4) and (5).

$$\begin{cases} \mathbf{Q}_1 = \begin{bmatrix} \cos \delta_f & \sin \delta_f & -d \cos \delta_f + l_f \sin \delta_f \\ \cos \delta_f & \sin \delta_f & d \cos \delta_f + l_f \sin \delta_f \\ \cos \delta_r & \sin \delta_r & -d \cos \delta_r - l_r \sin \delta_r \\ \cos \delta_r & \sin \delta_r & d \cos \delta_r - l_r \sin \delta_r \end{bmatrix}^T \\ \mathbf{Q}_2 = \begin{bmatrix} -\sin \delta_f & \cos \delta_f & l_f \cos \delta_f + d \sin \delta_f \\ -\sin \delta_f & \cos \delta_f & l_f \cos \delta_f - d \sin \delta_f \\ -\sin \delta_r & \cos \delta_r & -l_r \cos \delta_r + d \sin \delta_r \\ -\sin \delta_r & \cos \delta_r & -l_r \cos \delta_r - d \sin \delta_r \end{bmatrix}^T \end{cases} \quad (4)$$

$$\begin{cases} \mathbf{F}_x = [F_{x11} & F_{x12} & F_{x21} & F_{x22}]^T \\ \mathbf{F}_y = [F_{y11} & F_{y12} & F_{y21} & F_{y22}]^T \end{cases} \quad (5)$$

where, the tire longitudinal force and lateral force can be expressed as functions of the tire vertical force F_{zij} , the slip ratio s_{ij} and the tire slip angle α_{ij} and the road adhesion coefficient μ_{ij} [10].

$$F_{xij} = f_x(F_{zij}, s_{ij}, \alpha_{ij}, \mu_{ij}), F_{yij} = f_y(F_{zij}, s_{ij}, \alpha_{ij}, \mu_{ij}) \quad (6)$$

Tire vertical force F_{zij} , which has a significant influence on tire force, can be determined by using the method proposed by [42].

$$\begin{cases} F_{zfl} = \frac{m_v g l_r}{2l} - \frac{m_v h_g a_x}{2l} - \frac{m_v h_g a_y}{ld} + \frac{m_v a_x a_y h_g^2}{gdl} \\ F_{zfr} = \frac{m_v g l_r}{2l} - \frac{m_v h_g a_x}{2l} + \frac{m_v h_g a_y}{ld} - \frac{m_v a_x a_y h_g^2}{gdl} \\ F_{zrl} = \frac{m_v g l_f}{2l} + \frac{m_v h_g a_x}{2l} - \frac{m_v h_g a_y}{ld} - \frac{m_v a_x a_y h_g^2}{gdl} \\ F_{zrr} = \frac{m_v g l_f}{2l} + \frac{m_v h_g a_x}{2l} + \frac{m_v h_g a_y}{ld} + \frac{m_v a_x a_y h_g^2}{gdl} \end{cases} \quad (7)$$

where, a_x , a_y , and h_g , represent longitudinal acceleration, lateral acceleration and height of C.G. of the autonomous vehicle, respectively. g is the acceleration due to gravity.

The angular speed of each wheel ω_{ij} can be obtained by driving torque T_{dij} or braking torque T_{bij} of the in-wheel motor based on wheel rolling dynamics.

$$I_w \dot{\omega}_{ij} = T_{dij} - T_{bij} - F_{xij} R_e - f F_{zij} R_e \quad (8)$$

where, the parameters I_w , R_e and f are the moment of inertia of the electric wheel, the effective radius of the wheel, and rolling resistance coefficient of tire-road, respectively. The slip ratio of each wheel s_{ij} can be calculated by the angular speed and the longitudinal speed of the center of each wheel v_{wij} , which can be obtained by

$$s_{ij} = \begin{cases} \frac{R_e \omega_{ij}}{v_{wij}} - 1, & v_{wij} > R_e \omega_{ij} \\ 1 - \frac{v_{wij}}{R_e \omega_{ij}}, & v_{wij} < R_e \omega_{ij} \end{cases} \quad (9)$$

Based on wheel dynamics analysis, the tire slip angle can be described by

$$\begin{cases} \alpha_{fl} = \delta_f - \tan^{-1}\left(\frac{v + l_f r}{u - dr}\right) & \alpha_{rl} = \delta_r - \tan^{-1}\left(\frac{v - l_r r}{u - dr}\right) \\ \alpha_{fr} = \delta_f - \tan^{-1}\left(\frac{v + l_f r}{u + dr}\right) & \alpha_{rr} = \delta_r - \tan^{-1}\left(\frac{v - l_r r}{u + dr}\right) \end{cases} \quad (10)$$

Many researchers have built multifarious tire models, such as Magic Formula [8] and the brush model, through experimental analyzing and data fitting. Magic Formula has the characteristics of high fitting accuracy, strong unity, and convenient programming and more accepted by researchers and tire manufactures [33].

$$F_p = D \sin\{\text{Carctan}[BX - E(BX - \arctan(BX))]\} \quad (11)$$

where, F_p denotes the longitudinal tire force F_{xp} or the lateral tire force F_{yp} , correspondingly when X represents tire slip ratio s or tire slip angle α under pure slip conditions. The parameters B , C , D , and E are the stiffness factor, shape factor, peak factor, and curvature factor, respectively. The peak factor D is related to the vertical tire load and the tire-road coefficient adhesion [43].

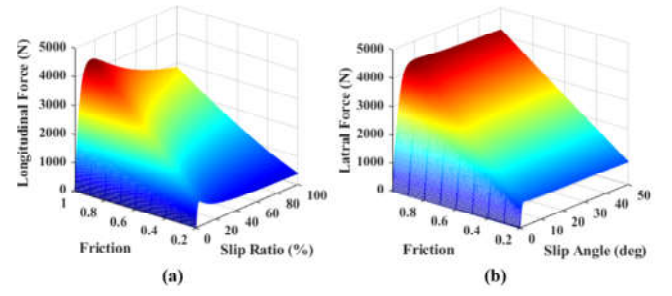


Fig. 3. Magic Formula tire model under pure slip conditions. (a) Longitudinal characteristics of the tire. (b) lateral characteristics of the tire.

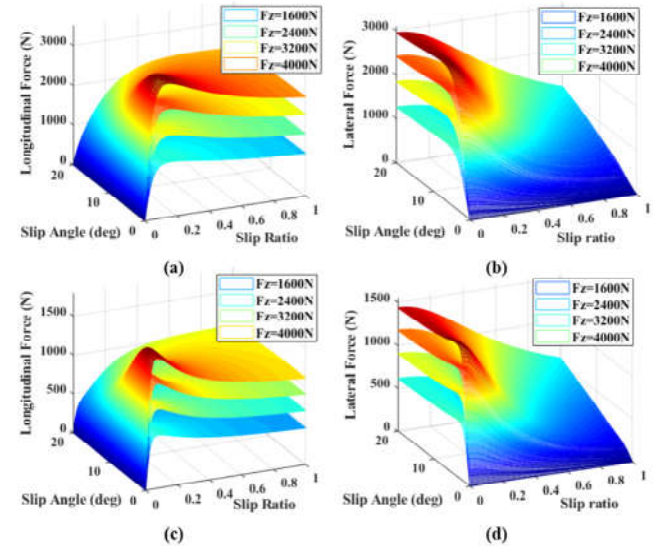


Fig. 4. Combined tire force change under $\mu = 0.4, 0.8$. (a) Tire longitudinal force characteristics ($\mu = 0.8$). (a) Tire longitudinal force characteristics ($\mu = 0.4$). (a) Tire lateral force characteristics ($\mu = 0.8$). (a) Tire lateral force characteristics ($\mu = 0.4$).

Fig. 3 shows the longitudinal/lateral characteristics of the tire under the tire vertical load 4000N. Fig. 3(a) shows the longitudinal force F_x increases sharply and linearly as the tire slip increases before 20%. If the slip ratio continues to rise after that value, the tire longitudinal force decreases continuously and nonlinearly from its peak value. The tire longitudinal force increases approximately proportional to the increase of the tire-road adhesion coefficient. Fig. 3(b) describes a similar phenomenon that the tire lateral force changes with the tire slip angle and the road adhesion coefficient. It is worth noting that the influence of the slip ratio on the longitudinal force is greater than the impact of the slip angle on the longitudinal force.

The tire model described above shows the tire force under pure longitudinal/side slip conditions. The tire longitudinal/lateral force interacts with each other in a combined slip condition. The tire force under combine conditions should be obtained because the characteristics of nonlinearity and coupling property of the tire are the basis for analyzing the feature of the nonlinear tire dynamics. The improved Magic Formula is further modified to describe the tire force under combine conditions. The combined tire force F_x/F_y is described by

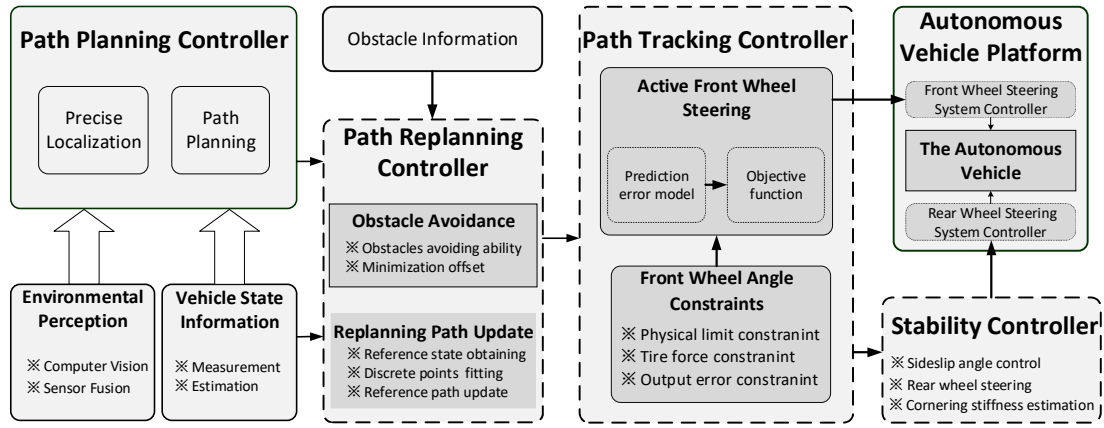


Fig. 5. The overall architecture of path tracking and handling stability control strategy with collision avoidance.

$$\begin{cases} F_x = \frac{\sigma_x}{\sqrt{\sigma_x^2 + \sigma_y^2}} F_{xp} \\ F_y = \frac{\sigma_y}{\sqrt{\sigma_x^2 + \sigma_y^2}} F_{yp} \end{cases} \begin{cases} \sigma_x = \frac{s}{1 + s} \\ \sigma_y = \frac{\tan \alpha}{1 + s} \end{cases} \quad (12)$$

where, F_{xp} and F_{yp} represent the pure tire longitudinal force and the pure tire lateral force, respectively. σ_x and σ_y , respectively, are the normalized slip ratio and the lateral slip angle.

Fig. 4 shows the combined tire force changing with the tire slip angle and the tire slip ratio under a series of the vertical tire forces under the tire-road adhesion coefficient $\mu = 0.4, 0.8$. In the pure longitudinal slip $\alpha = 0$ or pure side $s = 0$ condition, the changing characteristics of the tire force with slip ratio or sideslip angle are the same as that shown in Fig. 3. Under the combined conditions, Fig. 4 (a)-(b) show the tire longitudinal force characteristics vary with the road adhesion coefficient. The nonlinear tire force decreases when the road adhesion coefficient decreases. The tire longitudinal force keeps dropping when the sideslip angle α increases at fixed slip ratio s . Fig. 4 (c)-(d) show a similar phenomenon. Remarkably, the tire lateral force is more affected by the slip ratio than the longitudinal force by the sideslip angle.

III. PATH TRACKING AND HANDLING STABILITY CONTROL STRATEGY WITH COLLISION AVOIDANCE

The section mainly introduces the overall architecture of the path tracking and handling stability control strategy with collision avoidance proposed for the autonomous vehicle, as shown in Fig. 5.

The proposed novel control system includes a path replanning controller, a path tracking controller, and a stability controller, which aims to improve the performance in terms of path tracking and handling stability as well as collision avoidance under extreme conditions.

The path replanning controller generates a new collision-free path for collision avoidance circumstances under the extreme conditions. The basic idea of collision avoidance is to guarantee the rapidity and smoothness of the rescheduling path to ensure the driving safety of the vehicle, and allow temporary deviate

from the original planned path to best meet avoiding colliding obstacle under emergency conditions. The replanned path based on a polynomial real-time fitting function is sent to the path tracking controller as a new desired trajectory when the vehicle is avoiding obstacles.

A prediction error model in the path tracking controller is used to predict the lateral tracking offset and yaw angle error in the prediction horizon. Meanwhile, an objective function with considering multiple constraints, e.g., physical limitations of the wheel angle, the tire-road adhesion coefficient, and the path tracking error, is constructed to minimize the tracking error. The vehicle handling stability is guaranteed by vehicle dynamics constraints with the maximum allowable error range. The optimal steering angle of the wheel is sent to the stability controller as well as the front wheel steering system of the vehicle.

The stability controller applies the active steering angle of the rear wheel by regulating the sideslip angle of C.G because the sideslip angle of C.G has a significant influence on the path tracking and handling stability of the vehicle under extreme conditions. The active steering of the rear wheel based on zero sideslip angle of C.G is utilized to prevent the car from excessive sideslip angle of C.G. In addition, the robustness of the control strategy for guaranteeing handling stability is improved account for estimating the tire cornering stiffness.

A. Path replanning controller

Collision avoidance of the autonomous vehicle is a crucial subject and worth studying through integrated chassis system control to provide passengers with a safe driving environment. A collision avoidance objective function is proposed in the path re-planning controller to realize the purpose of avoiding collision automatically. Under the relatively high road friction coefficient, collision avoidance can be easily realized by controlling the tire lateral force.

However, the friction coefficient cannot provide tire forces to support a high lateral acceleration of the autonomous vehicle when the vehicle is driving on the relatively low adhesion road, e.g., the ice and snow road. The maximum tire force that a tire can provide should be limited to the friction circle, and the lateral offset of the new planned path should be considered for safety reasons.

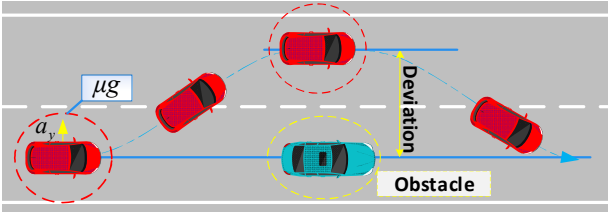


Fig. 6. The collision avoidance scenario of the autonomous vehicle.

Fig. 6 shows the collision avoidance scenario of the autonomous vehicle, which finds a collision-free path from the initial state to the target destination according to specific evaluation indicators, e.g., the shortest path offset and the least energy consumption principle. The lateral acceleration a_y of the vehicle maintains within the limit of the road adhesion circle.

In the path re-planning controller, the autonomous vehicle is regarded as a point mass model.

$$\begin{cases} \dot{v} = a_y \\ \dot{u} = 0 \\ \dot{\varphi} = \frac{a_y}{u} \end{cases} \quad (13)$$

Because the lateral acceleration must be bounded by a function of the tire-road friction coefficient, the lateral acceleration constraint of the vehicle is described as

$$|a_y| \leq \mu g \quad (14)$$

The system state of the vehicle can be obtained by (13) and (14).

$$\begin{cases} \dot{\mathbf{w}} = f(\mathbf{w}, \mathbf{v}) \\ \mathbf{v} \leq \mu g \end{cases} \quad (15)$$

where, $\mathbf{w} = [v \ u \ \varphi \ Y \ X]^T$ and $\mathbf{v} = a_y$ are the system state and the system input, respectively. The discrete state equation can be obtained by Taylor expansion at time t .

$$\mathbf{w}_{k+1,t} = \mathbf{g}_{k,t} \mathbf{w}_{k,t} + \mathbf{h}_{k,t} \mathbf{v}_{k,t} \quad (16)$$

where, $\mathbf{g}_{k,t} = \frac{\partial f}{\partial \mathbf{w}} T_c$ and $\mathbf{h}_{k,t} = \frac{\partial f}{\partial \mathbf{u}} T_c$ are system matrix and control matrix, with T_c denoting the calculating step length of the path replanning controller. $\frac{\partial f}{\partial \mathbf{w}}$ and $\frac{\partial f}{\partial \mathbf{u}}$ are the Jacobian matrices of $f(\mathbf{w}, \mathbf{v})$ respect to $\mathbf{w}$ and $\mathbf{v}$.

Here, the output of the vehicle system is defined as

$$\mathbf{z}_{k+1,t} = \mathbf{l}_{k+1,t} \mathbf{w}_{k+1,t} \quad (17)$$

where, output vector $\mathbf{z} = [Y_p \ X_p]^T$ is the predicted position of the vehicle.

To generate a new reference trajectory, the collision avoidance objective function $J_{ob,t}$ is designed, which indicates the sum of the distance from the autonomous vehicle to the obstacle position $\mathbf{z}_{o,k+1,t}$ at each predicted step in the prediction horizon H_p at time t .

$$J_{ob,t} = (\bar{\mathbf{z}}_t - \bar{\mathbf{z}}_{o,t})^T \mathbf{B}_m (\bar{\mathbf{z}}_t - \bar{\mathbf{z}}_{o,t}) \quad (18)$$

$$\bar{\mathbf{z}}_t = \begin{bmatrix} \mathbf{z}_{k+1,t} \\ \mathbf{z}_{k+2,t} \\ \vdots \\ \mathbf{z}_{k+H_p-1,t} \end{bmatrix}, \bar{\mathbf{z}}_{o,t} = \begin{bmatrix} \mathbf{z}_{o,k+1,t} \\ \mathbf{z}_{o,k+2,t} \\ \vdots \\ \mathbf{z}_{o,k+H_p-1,t} \end{bmatrix} \quad (19)$$

where, the state output of the vehicle $\bar{\mathbf{z}}_t$ in the prediction horizon at time t can be obtained by iteration of (17). The position of an obstacle $\bar{\mathbf{z}}_{o,t}$ is defined as the same dimension vector in the prediction horizon. The weight matrix $\mathbf{B}_m$ is

designed to keep the predicted trajectory of the vehicle in a safe route outside of an elliptical enclosure obstacle.

Not only should autonomous vehicles successfully avoid obstacles, but also the minimal lateral offset of an updated replanning trajectory should be simultaneously considered in the path replanning controller. Under the limit condition, the lateral acceleration of the vehicle should limit the maximum allowable lateral acceleration confined by vehicle dynamics and adhesion coefficient. To avoid obstacles, minimize lateral displacement and tire adhesion utilization, and achieve the purpose of vehicle stability when the path replanning controller generate a new reference trajectory, the cost function $J_{o,t}$ in the prediction horizon is described by

$$\begin{cases} J_{o,t} = (\bar{\mathbf{z}}_t - \bar{\mathbf{z}}_{l,t})^T \mathbf{Q}_0 (\bar{\mathbf{z}}_t - \bar{\mathbf{z}}_{l,t}) + \mathbf{v}_{l,t}^T \mathbf{P}_v \mathbf{v}_{l,t} + \chi_t J_{ob,t} \\ \text{s.t. } \mathbf{v}_{l,min,t} \leq \mathbf{v}_{l,t} \leq \mathbf{v}_{l,max,t} \end{cases} \quad (20)$$

with,

$$\bar{\mathbf{z}}_{l,t} = \begin{bmatrix} \mathbf{z}_{l,k+1,t} \\ \mathbf{z}_{l,k+2,t} \\ \vdots \\ \mathbf{z}_{l,k+H_p-1,t} \end{bmatrix}, \mathbf{v}_{l,t} = \begin{bmatrix} \mathbf{v}_{l,k+1,t} \\ \mathbf{v}_{l,k+2,t} \\ \vdots \\ \mathbf{v}_{l,k+H_p-1,t} \end{bmatrix} \quad (21)$$

where, $\bar{\mathbf{z}}_{l,t}$ and $\mathbf{v}_{l,t}$ are the locally desired trajectory and the local state input of the autonomous vehicle in the prediction horizon. $\mathbf{Q}_0$ and $\mathbf{P}_v$ are weight matrix of $\bar{\mathbf{z}}_{l,t}$ and $\mathbf{v}_{l,t}$. $\mathbf{v}_{l,min,t}$ and $\mathbf{v}_{l,max,t}$ represent the upper and lower limits of $\mathbf{v}_{l,t}$ in the prediction horizon. The extent factor χ_t represents the weight of avoiding obstacles. When χ_t is relatively large, it means the autonomous vehicle has a strong ability to avoid obstacles. However, it is also necessary to take it into account the lateral offset between the re-planned path and the global pathed path, as well as the stability of the vehicle under the limit condition by adjusting the weight coefficients $\mathbf{Q}_0$ and $\mathbf{P}_v$.

The optimal system control input $\mathbf{v}_{l,t}^*$ is obtained by using an active set algorithm in the control horizon H_C with the guaranteed computational efficiency. A new state equation can be updated by introducing $\mathbf{v}_{l,k+1,t}^*$ into (16) with the assumption of $\mathbf{w}_{l,k,t} = \mathbf{w}_{k,t}$.

$$\mathbf{w}_{l,k+1,t} = \mathbf{g}_{k,t} \mathbf{w}_{l,k,t} + \mathbf{h}_{k,t} \mathbf{v}_{l,k+1,t}^* \quad (22)$$

The information of predicted re-planned path at time t can be obtained according to (22).

$$\begin{cases} \mathbf{y}_{l,k+1,t} = \mathbf{C}_{k+1,t} \mathbf{w}_{l,k+1,t} \\ \mathbf{z}_{l,k+1,t} = \mathbf{E}_{k+1,t} \mathbf{w}_{l,k+1,t} \end{cases} \quad (23)$$

where, $\mathbf{y}_l = [\varphi_l \ Y_l]^T$ and $\mathbf{z}_l = X_l$, respectively, denote the local reference output of the restructuring route for safety reasons. $\mathbf{C}_t$ and $\mathbf{E}_t$ represent system output matrices related to $\mathbf{y}_l$ and $\mathbf{z}_l$, respectively.

$$\begin{cases} \mathbf{C}_t = [0 \ 0 \ 1 \ 1 \ 0] \\ \mathbf{E}_t = [0 \ 0 \ 0 \ 0 \ 1] \end{cases} \quad (24)$$

The predicted re-planned path is presented in the form of discrete points $\mathbf{y}_{l,k+1,t}/\mathbf{z}_{l,k+1,t}$. Those number increases with the extension of the prediction horizon.

Considering the control period and time delay of the data transmission from the path tracking controller to the path replanning controller, those discrete points obtained by (23) and (24) cannot be directly used as the new reference path in the path tracking controller. Therefore, curve fitting for discrete points provides an effective method to implement the smooth transmission of reference data and its application to the path tracking layer reasonably. The least-squares polynomial fitting

method is adopted to generate a dynamic matrix owing to high fitting accuracy and computational efficiency compared with other curve fitting methods, e.g., spline interpolation fitting and gauss fitting. ΔS represents the fitting degree of the polynomial curve in the prediction horizon.

$$\Delta S = (\bar{\mathbf{z}}_t \mathbf{g}_t - \bar{\mathbf{y}}_t)^T \mathbf{I}_n (\bar{\mathbf{z}}_t \mathbf{g}_t - \bar{\mathbf{y}}_t) \quad (25)$$

where,

$$\bar{\mathbf{y}}_t = \begin{bmatrix} \mathbf{y}_{k+1,t} \\ \mathbf{y}_{k+2,t} \\ \vdots \\ \mathbf{y}_{k+H_p-1,t} \end{bmatrix}, \bar{\mathbf{z}}_t = \begin{bmatrix} \mathbf{z}_{k+1,t}^n & \dots & \mathbf{z}_{k+1,t} & 1 \\ \mathbf{z}_{k+2,t}^n & \dots & \mathbf{z}_{k+2,t} & 1 \\ \vdots & \ddots & \vdots & \vdots \\ \mathbf{z}_{k+H_p-1,t}^n & \dots & \mathbf{z}_{k+H_p-1,t} & 1 \end{bmatrix} \quad (26)$$

with,

$$\mathbf{g} = \begin{bmatrix} \kappa_0 & \kappa_1 & \dots & \kappa_n \\ h_0 & h_1 & \dots & h_n \end{bmatrix} \quad (27)$$

where, $\mathbf{g}$ is the dynamic polynomial fitting matrix, which is determined by (25)-(27). The updated replanned path can be expressed as

$$\boldsymbol{\eta}_{r,t} = \mathbf{G}_g \mathbf{x}_t \quad (28)$$

where, $\boldsymbol{\eta}_r = [Y_r \ \varphi_r]$ is the replanned path in the prediction horizon after replanning due to collision avoidance in the globally planned path. $\mathbf{G}_g$ is the output matrix related to $\mathbf{g}$ and extends to the prediction horizon H_p . The replanned path can be updated in real-time, whose information can be sent to the vehicle path tracking controller in a way without time-consuming.

B. Path tracking controller

The path tracking controller aims to improve the path tracking performance of the autonomous vehicle by controlling the steering angle of the front wheel under extreme conditions. An objective function with multiple constraints is constructed for path tracking. The physical limit of the vehicle steering system and the adhesion coefficient limit are considered in multi-constraints. Besides, the maximum constraint of tracking error is also considered to ensure the vehicle's stability within the range of allowable tracking error under the limit condition.

Owing to the merits of the predicting future output and dynamic real-time rolling optimization, a predictive error model based on the MPC method is adopted to increase the anti-interference to the external environment and excessive deviation caused by the model mismatch [22]. Based on (1)-(12), the reference state $\mathbf{x}_r$, the reference control input $\mathbf{u}_r$ and the reference output $\mathbf{y}_r$ of the system are defined as

$$\begin{cases} \dot{\mathbf{x}}_r = \mathbf{f}(\mathbf{x}_r, \mathbf{u}_r) \\ \mathbf{y}_r = \mathbf{c} \mathbf{x}_r \end{cases} \quad (29)$$

Ignoring the higher-order terms for ensuring the calculation efficiency, Eq. (1) is expanded by Taylor series at the reference point $(\mathbf{x}_r, \mathbf{u}_r)$.

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}_r, \mathbf{u}_r) + \mathbf{a}(\mathbf{x} - \mathbf{x}_r) + \mathbf{b}(\mathbf{u} - \mathbf{u}_r) \quad (30)$$

where, $\mathbf{a} = \frac{\partial \mathbf{f}(\mathbf{x}, \mathbf{u})}{\partial \mathbf{x}} \bigg|_{\substack{\mathbf{x}=\mathbf{x}_r \\ \mathbf{u}=\mathbf{u}_r}}$ and $\mathbf{b} = \frac{\partial \mathbf{f}(\mathbf{x}, \mathbf{u})}{\partial \mathbf{u}} \bigg|_{\substack{\mathbf{x}=\mathbf{x}_r \\ \mathbf{u}=\mathbf{u}_r}}$, respectively, are

the Jacobian matrices of $\mathbf{f}(\mathbf{x}, \mathbf{u})$ respect to $\mathbf{x}$ and $\mathbf{u}$.

The goal of path tracking can be achieved by controlling the front wheel angle without controlling the rear wheel angle. The control input used in the path tracking controller can actually be described as $\mathbf{u} = [\delta_f \ 0]^T$ and $\mathbf{u}_r = [\delta_{fr} \ 0]^T$. The control of the steering angle of the rear wheel is considered in the stability controller.

The prediction error model can be obtained by subtracting (30) from (29).

$$\begin{cases} \dot{\tilde{\mathbf{x}}} = \mathbf{a}_t \cdot \tilde{\mathbf{x}} + \mathbf{b}_t \cdot \tilde{\mathbf{u}} \\ \tilde{\mathbf{y}} = \mathbf{c} \tilde{\mathbf{x}} \end{cases} \quad (31)$$

where, $\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{x}_r$, $\tilde{\mathbf{u}} = \mathbf{u} - \mathbf{u}_r$, $\tilde{\mathbf{y}} = \mathbf{y} - \mathbf{y}_r$.

The prediction error model (31) is a typical continuous state of the autonomous vehicle that cannot satisfy the discretization requirement of the vehicle controller. The prediction error model with discrete-time characteristics can be achieved by

$$\begin{cases} \tilde{\mathbf{x}}_{k+1,t} = \mathbf{a}_{k,t} \cdot \tilde{\mathbf{x}}_{k,t} + \mathbf{b}_{k,t} \cdot \tilde{\mathbf{u}}_{k,t} \\ \tilde{\mathbf{y}}_{k,t} = \mathbf{c}_{k,t} \cdot \tilde{\mathbf{x}}_{k,t} \end{cases} \quad (32)$$

where, $\mathbf{a}_{k,t} = \mathbf{I} + T \cdot \mathbf{a}_t$, $\mathbf{b}_{k,t} = T \cdot \mathbf{b}_t$. T and t denote the sampling period and the sampling time, respectively. The control input increment $\Delta \mathbf{u}_{k,t}$ in the sampling time t can be expressed by

$$\Delta \mathbf{u}_{k,t} = \tilde{\mathbf{u}}_{k,t} - \tilde{\mathbf{u}}_{k-1,t} \quad (33)$$

For more explicit expression of the prediction error model and the requirement of controller design convenient, the redefined state equation and output equation can be described according to (32) and (33).

$$\begin{cases} \tilde{\tilde{\mathbf{x}}}_{k+1,t} = \mathbf{A}_{k,t} \cdot \tilde{\tilde{\mathbf{x}}}_{k,t} + \mathbf{B}_{k,t} \cdot \Delta \mathbf{u}_{k,t} \\ \tilde{\tilde{\mathbf{y}}}_{k,t} = \mathbf{C}_{k,t} \cdot \tilde{\tilde{\mathbf{x}}}_{k,t} \end{cases} \quad (34)$$

where

$$\begin{cases} \tilde{\tilde{\mathbf{x}}}_{k,t} = \begin{bmatrix} \tilde{\mathbf{x}}_{k,t} \\ \tilde{\mathbf{u}}_{k-1,t} \end{bmatrix}, \mathbf{A}_{k,t} = \begin{bmatrix} \mathbf{a}_{k,t} & \mathbf{b}_{k,t} \\ \mathbf{0}_{\zeta_u \times \zeta_x} & \mathbf{I}_{\zeta_u} \end{bmatrix} \\ \mathbf{B}_{k,t} = \begin{bmatrix} \mathbf{b}_{k,t} \\ \mathbf{I}_{\zeta_u} \end{bmatrix}, \mathbf{C}_{k,t} = [\mathbf{c}_{k,t} \ \mathbf{0}_{\zeta_u}] \end{cases} \quad (35)$$

The subscript ζ_u and ζ_x represent the dimension of the state vector and control vector, respectively.

The objective function J_t is formulated to minimize output error in the prediction horizon N_p and minimize control input increment in the control horizon N_c . Two indicators in the objective function are considered where the first indicator denotes the effect of prediction output $\tilde{\tilde{\mathbf{y}}}_{k+i,t}$ tracking reference output $\tilde{\tilde{\mathbf{y}}}_{r,k+i,t}$ in the prediction horizon while the second indicator shows the performance of stationarity of variation of vehicle control inputs $\Delta \mathbf{u}_{k+i-1,t}$ in the control horizon.

$$J_t = \sum_{i=1}^{N_p} \|\tilde{\tilde{\mathbf{y}}}_{k+i,t} - \tilde{\tilde{\mathbf{y}}}_{r,k+i,t}\|_{\mathbf{q}_{\zeta_y \times \zeta_y}}^2 + \sum_{i=1}^{N_c} \|\Delta \mathbf{u}_{k+i-1,t}\|_{\mathbf{p}_{\zeta_u \times \zeta_u}}^2 \quad (36)$$

where, $\mathbf{q}_{\zeta_y \times \zeta_y}$ and $\mathbf{p}_{\zeta_u \times \zeta_u}$ are weight matrix of output error and control input increment, respectively. $\tilde{\tilde{\mathbf{y}}}_{r,k+i,t}$ is the reference error, which can be got by

$$\tilde{\tilde{\mathbf{y}}}_{r,k+i,t} = \boldsymbol{\eta}_{r,k+i,t} - \boldsymbol{\eta}_{r,k+i,t}, i = 1, \dots, N_p. \quad (37)$$

The prediction state $\tilde{\tilde{\mathbf{x}}}_{k+1,t}$ in the prediction horizon can be calculated by (34) and (35).

$$\begin{cases} \tilde{\tilde{\mathbf{x}}}_{k+1,t} = \bar{\mathbf{A}}_{k,t} \cdot \tilde{\tilde{\mathbf{x}}}_{k,t} + \bar{\mathbf{B}}_{k,t} \cdot \Delta \mathbf{u}_{k,t} \\ \tilde{\tilde{\mathbf{x}}}_{k+1,t} = \begin{bmatrix} \tilde{\tilde{\mathbf{x}}}_{k+1,t} \\ \tilde{\tilde{\mathbf{x}}}_{k+2,t} \\ \vdots \\ \tilde{\tilde{\mathbf{x}}}_{k+N_p-1,t} \end{bmatrix}, \Delta \mathbf{u}_{k,t} = \begin{bmatrix} \Delta \mathbf{u}_{k,t} \\ \Delta \mathbf{u}_{k+1,t} \\ \vdots \\ \Delta \mathbf{u}_{k+N_c-1,t} \end{bmatrix} \end{cases} \quad (38)$$

where, $\tilde{\tilde{\mathbf{x}}}_{k+1,t}$ and $\Delta \mathbf{u}_{k,t}$ represent prediction states and prediction inputs in the sampling time. $\bar{\mathbf{A}}_{k,t}$ and $\bar{\mathbf{B}}_{k,t}$ are the system matrix and control matrix state in the prediction horizon.

$$\begin{cases} \bar{\mathbf{A}}_{k,t} = \begin{bmatrix} \mathbf{A}_{k,t} \\ \mathbf{A}_{k,t}\mathbf{A}_{k+1,t} \\ \vdots \\ \mathbf{A}_{\alpha(k,0),t} \end{bmatrix} \\ \bar{\mathbf{B}}_{k,t} = \begin{bmatrix} \mathbf{B}_{k,t} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{A}_{k+1,t}\mathbf{B}_{k,t} & \mathbf{B}_{k+1,t} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{A}_{\alpha(k,1),t}\mathbf{B}_{k,t} & \mathbf{A}_{\alpha(k,2),t}\mathbf{B}_{k+1,t} & \dots & \mathbf{B}_{k+N_p-1,t} \end{bmatrix} \end{cases} \quad (39)$$

where, $\mathbf{A}_{\alpha(k,j),t}$ can be expressed by $\mathbf{A}_{(k+i),t}$.

$$\mathbf{A}_{\alpha(k,j),t} = \prod_{i=j}^{N_p-1} \mathbf{A}_{(k+i),t} \quad (40)$$

The predictive output $\bar{\boldsymbol{\eta}}_{(k+1),t}$ of the vehicle system can be obtained according to (32) and (38).

$$\bar{\boldsymbol{\eta}}_{k+1,t} = \boldsymbol{\Gamma}_{k,t} \cdot \tilde{\boldsymbol{\xi}}_{k,t} + \boldsymbol{\Theta}_{k,t} \cdot \Delta \mathbf{U}_{k,t} \quad (41)$$

where,

$$\boldsymbol{\Gamma}_{k,t} = \mathbf{C}_{k,t} \bar{\mathbf{A}}_{k,t}, \boldsymbol{\Theta}_{k,t} = \mathbf{C}_{k,t} \bar{\mathbf{B}}_{k,t}, \bar{\boldsymbol{\eta}}_{k+1,t} = \begin{bmatrix} \tilde{\boldsymbol{\eta}}_{k+1,t} \\ \tilde{\boldsymbol{\eta}}_{k+2,t} \\ \vdots \\ \tilde{\boldsymbol{\eta}}_{k+N_p-1,t} \end{bmatrix} \quad (42)$$

The objective function can be rewritten as

$$J_t(\Delta \mathbf{U}_{k,t}, \tilde{\boldsymbol{\xi}}_{k,t}) = (\bar{\boldsymbol{\eta}}_{(k+1),t} - \bar{\boldsymbol{\eta}}_{r(k+1),t})^T \mathbf{Q}(\bar{\boldsymbol{\eta}}_{(k+1),t} - \bar{\boldsymbol{\eta}}_{r(k+1),t}) + \Delta \mathbf{U}_{k,t}^T \mathbf{P} \Delta \mathbf{U}_{k,t} \quad (43)$$

where,

$$\begin{cases} \bar{\boldsymbol{\eta}}_{r,k+1,t} = \begin{bmatrix} \boldsymbol{\eta}_{k+1,t} \\ \boldsymbol{\eta}_{k+2,t} \\ \vdots \\ \boldsymbol{\eta}_{k+N_p-1,t} \end{bmatrix} - \begin{bmatrix} \boldsymbol{\eta}_{r,k+1,t} \\ \boldsymbol{\eta}_{r,k+2,t} \\ \vdots \\ \boldsymbol{\eta}_{r,k+N_p-1,t} \end{bmatrix} \\ \mathbf{Q} = \begin{bmatrix} \mathbf{q}_{\zeta_y \times \zeta_y} & 0 & \dots & 0 \\ 0 & \mathbf{q}_{\zeta_y \times \zeta_y} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \mathbf{q}_{\zeta_y \times \zeta_y} \end{bmatrix}_{\zeta_y N_p \times \zeta_y N_p} \\ \mathbf{P} = \begin{bmatrix} \mathbf{p}_{\zeta_u \times \zeta_c} & 0 & \dots & 0 \\ 0 & \mathbf{p}_{\zeta_u \times \zeta_c} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \mathbf{p}_{\zeta_u \times \zeta_c} \end{bmatrix}_{\zeta_u N_c \times \zeta_u N_c} \end{cases} \quad (44)$$

The objective function (43) can be transformed into standard quadratic programming (QP) form for convenience mathematical computing, which can be expressed as

$$J_t = \Delta \mathbf{U}_{k,t}^T \mathbf{H}_t \Delta \mathbf{U}_{k,t} + \mathbf{f}_t^T \Delta \mathbf{U}_{k,t} + \mathbf{d}_{k,t} \quad (45)$$

where,

$$\begin{cases} \mathbf{H}_t = \boldsymbol{\Theta}_{k,t}^T \mathbf{Q} \boldsymbol{\Theta}_{k,t} + \mathbf{P} \\ \mathbf{f}_t = 2(\boldsymbol{\Gamma}_{k,t} \cdot \tilde{\boldsymbol{\xi}}_{k,t} - \bar{\boldsymbol{\eta}}_{r(k+1),t})^T \mathbf{Q} \boldsymbol{\Theta}_{k,t} \end{cases} \quad (46)$$

The quadratic $\mathbf{H}_t$ represents symmetric matrix and must be positive definite while vector $\mathbf{f}_t$ shows the linear term in the optimization function. $\mathbf{d}_{k,t}$, as the remainder term in the objective function, is ignored because the system input increment $\Delta \mathbf{U}_{k,t}$ is independent with $\mathbf{d}_{k,t}$.

The comprehensive constraints, e.g., the restriction of the physical limitations of the wheel angle, tire-road adhesion coefficient, and path tracking error, have to take it into account in QP because the vehicle must meet the demand of the vehicle dynamics, especially in the extreme condition. The constraints of the wheel angle of the autonomous vehicle can be described by

$$\mathbf{u}_{min} \leq \mathbf{u}_{k,t} \leq \mathbf{u}_{max} \quad (47)$$

where, $\mathbf{u}_{max}$ and $\mathbf{u}_{min}$ respectively are the maximum bound and minimum bound of the steering angles, indicating the extreme steering angles that the left and right side of the wheel. It mainly depends on physical limit constraint and tire force constraint.

$$\begin{cases} \mathbf{u}_{max} = \min(\mathbf{u}_{max1}, \mathbf{u}_{max2}) \\ \mathbf{u}_{min} = \max(\mathbf{u}_{min1}, \mathbf{u}_{min2}) \end{cases} \quad (48)$$

where, $\mathbf{u}_{min1}$ and $\mathbf{u}_{max1}$ are the maximum bound and the minimum bound of the structural physical constraints of the steering systems. $\mathbf{u}_{max2} = [\delta_{f,max} \ 0]^T$ and $\mathbf{u}_{min2} = [-\delta_{f,max} \ 0]^T$ are the constraints of the wheel steering angle related to the road adhesion coefficient. The adhesion coefficient and vehicle motion are considered in the form of the maximum steering angle of the wheel [21].

$$\begin{cases} \delta_{f,max} = \arctan\left(\frac{(l_f + l_r)\mu g}{u^2} - \tan(\alpha_{sl,r})\right) + \alpha_{sl,f} \\ \alpha_{sl,f} = \arctan\frac{3\mu F_z}{C_f} \end{cases} \quad (49)$$

To convert constraints into $\Delta \mathbf{U}_{k,t}$, the constraints of output vector $\bar{\mathbf{u}}_t$ are designed as

$$\bar{\mathbf{u}}_{min,t} - \bar{\mathbf{u}}_{r,t} \leq \bar{\mathbf{u}}_t \leq \bar{\mathbf{u}}_{max,t} - \bar{\mathbf{u}}_{r,t} \quad (50)$$

where,

$$\begin{cases} \bar{\mathbf{u}}_{min,t} = \begin{bmatrix} \mathbf{u}_{min,k,t} \\ \mathbf{u}_{min,k+1,t} \\ \vdots \\ \mathbf{u}_{min,k+N_c-1,t} \end{bmatrix}, \bar{\mathbf{u}}_{r,t} = \begin{bmatrix} \mathbf{u}_{r,k,t} \\ \mathbf{u}_{r,k+1,t} \\ \vdots \\ \mathbf{u}_{r,k+N_c-1,t} \end{bmatrix} \\ \bar{\mathbf{u}}_{max,t} = \begin{bmatrix} \mathbf{u}_{max,k,t} \\ \mathbf{u}_{max,k+1,t} \\ \vdots \\ \mathbf{u}_{max,k+N_c-1,t} \end{bmatrix}, \bar{\mathbf{u}}_t = \begin{bmatrix} \tilde{\mathbf{u}}_{k,t} \\ \tilde{\mathbf{u}}_{k+1,t} \\ \vdots \\ \tilde{\mathbf{u}}_{k+N_c-1,t} \end{bmatrix} \end{cases} \quad (51)$$

The output vector $\bar{\mathbf{u}}_t$ can be determined by (33) and (51).

$$\bar{\mathbf{u}}_t = \mathbf{N}_1 \cdot \tilde{\mathbf{u}}_{k,t} + \mathbf{N}_2 \Delta \mathbf{U}_{k,t} \quad (52)$$

where,

$$\mathbf{N}_1 = \begin{bmatrix} \mathbf{I}_{\zeta_u} \\ \mathbf{I}_{\zeta_u} \\ \vdots \\ \mathbf{I}_{\zeta_u} \end{bmatrix}_{\zeta_u N_c \times \zeta_u}, \mathbf{N}_2 = \begin{bmatrix} \mathbf{I}_{\zeta_u} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{I}_{\zeta_u} & \mathbf{I}_{\zeta_u} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{I}_{\zeta_u} & \mathbf{I}_{\zeta_u} & \dots & \mathbf{I}_{\zeta_u} \end{bmatrix}_{\zeta_u N_c \times \zeta_u N_c} \quad (53)$$

The constraint of $\Delta \mathbf{U}_{k,t}$ is described as a canonical form according to (50)-(53).

$$\begin{bmatrix} \mathbf{N}_2 \\ -\mathbf{N}_2 \end{bmatrix} \Delta \mathbf{U}_{k,t} \leq \begin{bmatrix} \bar{\mathbf{u}}_{max,t} - \bar{\mathbf{u}}_{r,t} \\ -(\bar{\mathbf{u}}_{min,t} - \bar{\mathbf{u}}_{r,t}) \end{bmatrix} \quad (54)$$

In order to coordinate the path tracking and handling stability of the autonomous vehicle, the stability margin and path tracking error are expressed by the system output constraints. The predictive output meets the maximum lateral offset after the stability of the vehicle is given the priority.

$$\bar{\boldsymbol{\eta}}_{min,t} - \bar{\boldsymbol{\eta}}_{r,t} \leq \bar{\boldsymbol{\eta}}_t \leq \bar{\boldsymbol{\eta}}_{max,t} - \bar{\boldsymbol{\eta}}_{r,t} \quad (55)$$

where, $\bar{\boldsymbol{\eta}}_{min,t}$ and $\bar{\boldsymbol{\eta}}_{max,t}$ are the minimum and maximum limits in the prediction, respectively.

$$\bar{\boldsymbol{\eta}}_{min,t} = \begin{bmatrix} \boldsymbol{\eta}_{min,t} \\ \boldsymbol{\eta}_{min,t} \\ \vdots \\ \boldsymbol{\eta}_{min,t} \end{bmatrix}, \bar{\boldsymbol{\eta}}_{max,t} = \begin{bmatrix} \boldsymbol{\eta}_{max,t} \\ \boldsymbol{\eta}_{max,t} \\ \vdots \\ \boldsymbol{\eta}_{max,t} \end{bmatrix}, \bar{\boldsymbol{\eta}}_{r,t} = \begin{bmatrix} \boldsymbol{\eta}_{r,t} \\ \boldsymbol{\eta}_{r,t} \\ \vdots \\ \boldsymbol{\eta}_{r,t} \end{bmatrix} \quad (56)$$

The following inequality constraints can be obtained by (41) and (54).

$$\bar{\boldsymbol{\eta}}_{min,t} - \bar{\boldsymbol{\eta}}_{r,t} \leq \boldsymbol{\Gamma}_{k,t} \cdot \tilde{\boldsymbol{\xi}}_t + \boldsymbol{\Theta}_{k,t} \cdot \Delta \mathbf{U}_{k,t} \leq \bar{\boldsymbol{\eta}}_{max,t} - \bar{\boldsymbol{\eta}}_{r,t} \quad (57)$$

The constraint of the system control increment $\Delta \mathbf{U}_{k,t}$ can be got based on (56).

$$\begin{bmatrix} \mathbf{\Theta}_{k,t} \\ -\mathbf{\Theta}_{k,t} \end{bmatrix} \Delta \mathbf{U}_{k,t} \leq \begin{bmatrix} \bar{\eta}_{max,t} - \bar{\eta}_{r,t} - \mathbf{r}_{k,t} \cdot \tilde{\xi}_t \\ -(\bar{\eta}_{min,t} - \bar{\eta}_{r,t} - \mathbf{r}_{k,t} \cdot \tilde{\xi}_t) \end{bmatrix} \quad (58)$$

The multiple constraints of input increment can be given in a canonical form, according to (54) and (58).

$$\begin{bmatrix} \mathbf{N}_2 \\ -\mathbf{N}_2 \\ \mathbf{\Theta}_{k,t} \\ -\mathbf{\Theta}_{k,t} \end{bmatrix} \Delta \mathbf{U}_{k,t} \leq \begin{bmatrix} \bar{\mathbf{u}}_{max,t} - \bar{\mathbf{u}}_{r,t} \\ -(\bar{\mathbf{u}}_{min,t} - \bar{\mathbf{u}}_{r,t}) \\ \bar{\eta}_{max,t} - \bar{\eta}_{r,t} - \mathbf{r}_{k,t} \cdot \tilde{\xi}_t \\ -(\bar{\eta}_{min,t} - \bar{\eta}_{r,t} - \mathbf{r}_{k,t} \cdot \tilde{\xi}_t) \end{bmatrix} \quad (59)$$

The objective function (31) with inequality constraints (59) is the standard QP problem, which can be solved by the active-set method to ensure the calculation efficiency. The optimal solution sequence of control increment $\Delta \mathbf{U}_{k,t}^*$ in the control horizon can be got.

$$\Delta \mathbf{U}_{k,t}^* = [\Delta \mathbf{u}_{k,t}^* \quad \Delta \mathbf{u}_{k+1,t}^* \quad \cdots \quad \Delta \mathbf{u}_{k+N_c-1,t}^*]^T \quad (60)$$

where, $\Delta \mathbf{u}_{k+i,t}^*$ ($i = 0, 1, \dots, N_c - 1$) is the optimal solution at time $k + i$ in the control horizon. The first element of the optimal solution sequence, as the optimal control steering angle of the front wheel at the current time, is sent to the vehicle steering system. The control input $\mathbf{u}_{k,t}^*$ can be obtained by (60) and (33).

$$\mathbf{u}_{k,t}^* = \Delta \mathbf{u}_{k,t}^* + \mathbf{u}_{k-1,t} \quad (61)$$

The optimal control input of the autonomous vehicle can be obtained in sequence by repeating the numerical solution after entering the next control cycle.

C. Stability controller

The heading angle of the vehicle, directly related to the plane motion, is determined by the yaw angle and the sideslip angle of C.G [43]. In the path tracking controller, the yaw angle is a controlled state to achieve the vehicle path tracking. However, the risk of losing the stability of the autonomous vehicle increases due to excessive sideslip angle of C.G when the vehicle is in extreme conditions, e.g., high speed and low-adhesion road. The handling stability of the autonomous vehicle is of importance, even though the path tracking of the autonomous vehicle is guaranteed. A new method that the rear-wheel steering based on the zero sideslip angle of C.G is utilized in the stability controller to prevent the car from losing handling stability.

The two degrees of freedom vehicle model with varying tire cornering stiffness exhibits excellent real-time performance and strong engineering realization.

$$\begin{cases} m u (\dot{\beta} + r) = 2C_f \alpha_f + 2C_r \alpha_r \\ I_z \dot{r} = 2l_f C_f \alpha_f - 2l_r C_r \alpha_r \end{cases} \quad (62)$$

where, C_f and C_r are the estimated coaxial cornering stiffness of the front tire and the rear tire, respectively. The sideslip angle of C.G and the yaw rate r can be determined by our previous study [40]. α_f and α_r are respectively the front tire slip angle and the rear tire slip angle, which can be estimated by

$$\begin{cases} \alpha_f(k) = \delta_f(k-1) - \beta(k-1) - \frac{l_f r(k-1)}{u(k-1)} \\ \alpha_r(k) = \delta_r(k-1) - \beta(k-1) + \frac{l_r r(k-1)}{u(k-1)} \end{cases} \quad (63)$$

The Laplace transform $\beta(s)$ is obtained by (62) and (63).

$$\beta(s) = \frac{\begin{bmatrix} 2C_f \delta_f(s) + 2C_r \delta_r(s) & mu + \frac{2}{u}(C_f l_f - C_r l_r) \\ 2l_f C_f \delta_f(s) - 2l_r C_r \delta_r(s) & I_{zz} s + \frac{2}{u}(l_f^2 C_f + l_r^2 C_r) \end{bmatrix}}{\begin{bmatrix} mu + 2(C_f + C_r) & mu + \frac{2}{u}(C_f l_f - C_r l_r) \\ 2(l_f C_f - l_r C_r) & I_{zz} s + \frac{2}{u}(l_f^2 C_f + l_r^2 C_r) \end{bmatrix}} \quad (64)$$

where, $\delta_f(s)$ and $\delta_r(s)$ are the Laplace transform, respectively.

The reasonable controlling method of the rear-wheel steering angle based on the front wheel steering angle is designed to ensure the stability of the autonomous vehicle under extreme conditions. The rear-wheel steering angle is in proportion to the front wheel steering angle.

$$\delta_r(s) = k(s) \delta_f(s) \quad (65)$$

where, $k(s)$ is the proportion coefficient of the steering angle of the rear wheel relative to the steering angle of the front wheel. The transfer function of the sideslip angle of C.G to the steering angle of the front wheel is calculated according to (64) and (65).

$$\frac{\beta(s)}{\delta_f(s)} = \frac{\begin{bmatrix} 2C_f + 2C_r k(s) & mu + \frac{2}{u}(C_f l_f - C_r l_r) \\ 2l_f C_f - 2l_r C_r k(s) & I_{zz} s + \frac{2}{u}(l_f^2 C_f + l_r^2 C_r) \end{bmatrix}}{\begin{bmatrix} mu + 2(C_f + C_r) & mu + \frac{2}{u}(C_f l_f - C_r l_r) \\ 2(l_f C_f - l_r C_r) & I_{zz} s + \frac{2}{u}(l_f^2 C_f + l_r^2 C_r) \end{bmatrix}} \quad (66)$$

To prevent the vehicle from losing stability due to the excessive sideslip angle under the limit condition, and to ensure the performance of the path tracking of the autonomous vehicle, the zero sideslip angle of C.G is designed based on the rear-wheel steering control.

The response of sideslip angle to any steering angle of the front wheel in steady-state steering motion is 0 when the proportion coefficient is appropriately designed. The proportion coefficient can be determined by (67) and (68).

$$k(s) = \frac{k_0}{1 + T_e s} - \frac{C_f T_e s}{C_r (1 + T_e s)} \quad (67)$$

$$k_0 = -\frac{l_r \left(1 - \frac{m l_f}{2 l_r C_r} u^2\right)}{l_f \left(1 + \frac{m l_f}{2 l_r C_f} u^2\right)}, T_e = \frac{I_{zz} u}{2 l_l C_f + m l_r u^2} \quad (68)$$

Similarly, δ_r is restricted by the physical constraints of the structure and the limit constraint related to the road condition. The lateral motion of the autonomous vehicle is mainly determined by the yaw angle when the sideslip angle is controlled to zero. It does not affect the path tracking performance because the yaw angle determines the vehicle heading angle in the path tracking controller.

The cornering stiffness of the tire also is estimated based on the predefined Magic Formula model to improve the robustness of the autonomous vehicle under extreme conditions. The coaxial tire cornering stiffness C_f and C_r are approximately the average cornering stiffness of the tire C_{ij} on the assumption of the 2DOF model of the vehicle being a monorail model.

$$C_f = \frac{C_{fl} + C_{fr}}{2}, C_r = \frac{C_{rl} + C_{rr}}{2} \quad (69)$$

The tire cornering stiffness is greatly influenced by vehicle motion and road condition whose value can be decreased to almost zero because of the tire force saturation under low adhesion road.

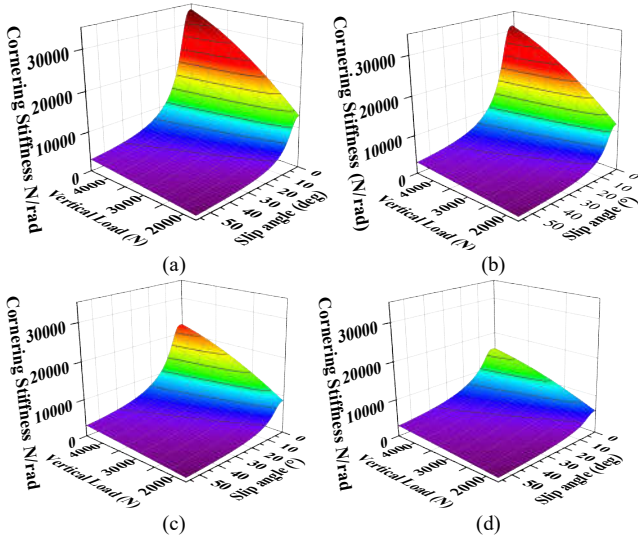


Fig. 7. Nominal tire cornering stiffness under tire slip ratios. (a) Tire slip ratio (0). (b) Tire slip ratio (0.1). (c) Tire slip ratio (0.2). (d) Tire slip ratio (0.5).

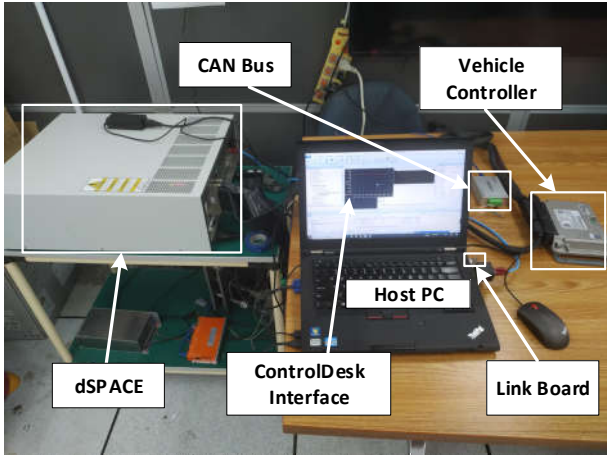


Fig. 8. HIL verification platform.

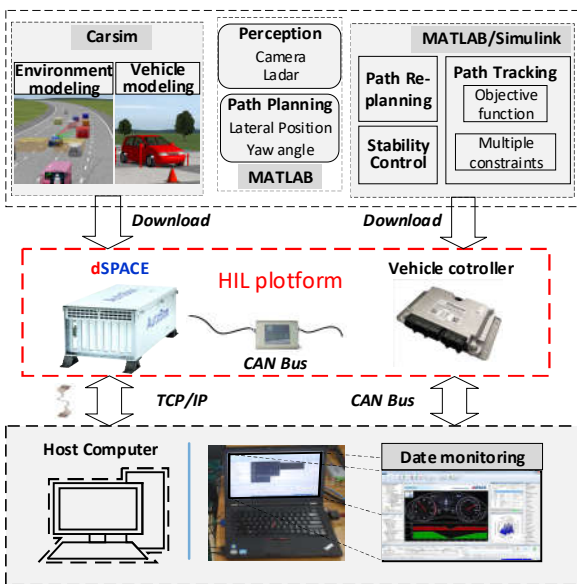


Fig. 9. The structure of the implementation of HIL platform.

To improve robustness, the tire cornering stiffness is estimated by nominal tire cornering stiffness C_n , which can be obtained from differentiating F_y against the tire lateral slip angle, according to (11).

$$\begin{cases} C_n = \frac{\partial F_y}{\partial \alpha} = \frac{f_{\alpha 1} \cos[C \tan^{-1}(f_{\alpha 2})]}{1 + (f_{\alpha 2})^2} \\ f_{\alpha 1} = DCB \left[1 - E + \frac{E}{(B\alpha)^2 + 1} \right] \\ f_{\alpha 2} = B\alpha - E[B\alpha - \tan^{-1}(B\alpha)] \end{cases} \quad (70)$$

where, $f_{\alpha 1}$ and $f_{\alpha 2}$ are the functions of α .

Fig. 7 shows the nominal cornering stiffness characteristic changes with the tire vertical load and the tire slip angle under the tire slip ratio $s = 0, 0.1, 0.2$, and 0.5 . The nominal cornering stiffness of the tire remains almost unchanged when the slip angle of the tire is less than 6° . It decreases to a specific value with the increase of the tire slip angle in each vertical load. The tire cornering stiffness increases with the tire vertical load increasing at a fixed slip angle. With the tire slip ratio increasing, the nominal cornering stiffness decreases at the same vertical load and the slip angle of the tire. The smaller the sideslip angle is, the more significant decrease is.

The actual tire lateral stiffness C_{ij} is assumed to exist a positive proportional relationship to the nominal cornering stiffness.

$$C_{ij} = \rho_{ij}(k) C_{n,ij}, (i = f, l; j = l, r) \quad (71)$$

where, ρ_{ij} is the utilization ratio of tire longitudinal force. ρ_{ij} can be estimated by

$$\rho_{ij}(k) = \frac{F_{xij}(k-1)}{F_{zij}(k-1)}, (i = f, l; j = l, r) \quad (72)$$

IV. HIL SIMULATIONS

A. HIL simulation platform

In ‘V’ diagram in the development process of the control system, the real-time simulation environment HIL is essential for the development of the vehicle control system after verification in the model in the loop simulation [44]. As shown in Fig. 8, the HIL verification platform was constructed to verify the real-time and effectiveness of the proposed control algorithm.

Fig. 9 displays the implementation process of HIL platform. The environment model and the nonlinear dynamic model of the autonomous vehicle, whose structural parameters refer to our previous work [41], were built in high-fidelity simulation software Carsim. Then, the models constructed in Carsim were downloaded in dSPACE/Autobox with a DS1006 processing board though Real-Time Interface (RTI) provided by dSPACE. The proposed control algorithm, built-in MATLAB/Simulink, was downloaded in the vehicle controller in the form of C code by Real Time Workspace (RTW) provided by MATLAB/Simulink. CAN bus is used to communicate information between dSPACE/Autobox and the vehicle controller. After obtaining the vehicle states’ information, the vehicle controller subsequently sends the control commands (The steering wheel angles) to dSPACE/Autobox. ControlDesk interface in host computer can be used to monitor vehicle states and adjust parameters through TCP/IP or Can-Bus. The control parameters are listed in Table I.

TABLE I
PARAMETERS OF CONTROLLER

Parameters	Description	Value
N_P	Prediction horizon	20
N_C	Control horizon	5
q	Weighting matrices	diag(200, 100)
r	Weighting matrices	500eye(10)
Y_{mas}	Tracking error margin	0.5
χ_t	Extent factor	1000
T	Sampling period	0.04
Q_0	Weighting matrices	100eye(20)

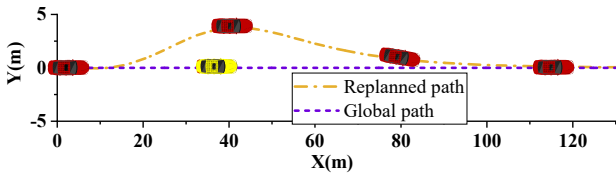


Fig. 10. HIL simulation results of replanning trajectory for collision avoidance.

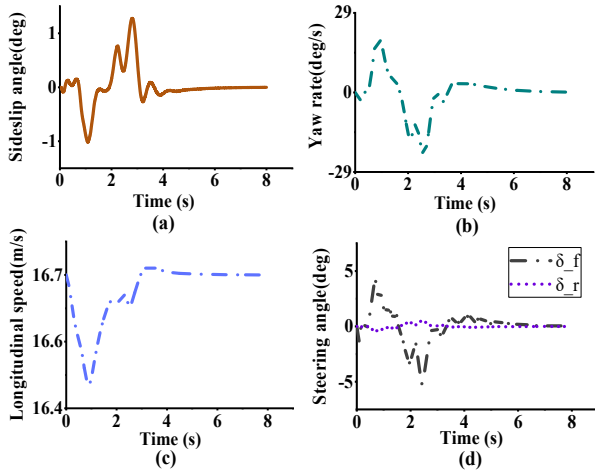


Fig. 11. HIL simulation results of vehicle states. (a). Sideslip angle of C.G. (b). Yaw rate. (c). Longitudinal speed. (d). Steering angle.

B. Simulation results

A specific test scenario *Case 1* that the detected obstacle is preset artificially in front of the vehicle on the global planned path is utilized to demonstrate the collision avoidance ability of the autonomous vehicle.

To verify the effectiveness of the proposed control method, two test scenarios *Case 2* and *Case 3* are implemented for the performance of path tracking and handling stability under extreme scenarios. *Case 2* mainly verifies the performance of the handling stability and path tracking when the vehicle is driving at high speed while *Case 3* checks the control strategy when the vehicle operates on a low adhesion road. Two other control methods are compared in the same experiment test scenarios. The computation time in two scenarios, *Case 2* and *Case 3*, is also analyzed to investigate the real-time implementation of the controller. The real-time computation time ability reflects the performance of the computation capability of the control algorithm.

One of the comparative control methods is the control method that only controls the steering angle of the front wheel without controlling the rear-wheel steering angle. The other control method simultaneously and independently controls the steering angle of the front wheel and the rear wheel based on an MPC method. To analyze the HIL simulation results conveniently and concisely, the two comparative control methods are marked as 'C1' and 'C2', respectively. The novel control strategy presented in this paper is marked as 'C0'.

Case 1: Collision avoidance

In order to verify the performance of collision avoidance of the autonomous vehicle, a virtual scenario to simulate the real scene is designed. It can reflect the ability to avoid obstacles and maintain the handling stability of the autonomous vehicle under the low tire-road adhesion coefficient from the perspective of vehicle state and the replanned path.

In the typical scene, the road is set as ice and snow road with the tire-road adhesion coefficient $\mu = 0.3$. The autonomous vehicle is driving along the planned path, and the speed of the car is 16.7m/s. The planned path is similar to a straight lane on a highway. There exists a predefined stationary obstacle (width 2m) in advance in front of the vehicle on the global planned path. HIL simulation results for collision avoidance of the autonomous vehicle under the low tire-road adhesion coefficient are shown in Fig. 10, and Fig. 11.

Fig. 10 shows the path replanning of the autonomous vehicle when encountering an obstacle on the global planned path. The purple dotted line is the globally planned path set in advance in MATLAB, which simulates the planned path obtained by the path planning controller. The yellow vehicle on the global planned path imitates the obstacle, whose information can be got through millimeter-wave radar and onboard video camera in the analog receiver. The yellow dash-dot is the replanned path after the obstacle is detected. The series of red vehicles are the locations of the autonomous vehicle at different times from the beginning to the end of obstacle avoiding trajectory.

As shown in Fig. 10, a new replanned path is regenerated and deviates from the original planned path when the autonomous vehicle detects an obstacle in front of it. The maximum deviation between the planned path and the replanned path is 3.8m. The replanned trajectory is a smooth polynomial curve, which ensures the stability of the vehicle. Furthermore, the autonomous vehicle can return to the global planned path after successful obstacle avoidance.

Fig. 11(a)-(b) show the results of the sideslip angle of C.G and the yaw rate when the autonomous vehicle avoids obstacles on the low adhesion road. When the vehicle searches for the new accessible replanning path, the sideslip angle of C.G and yaw rate of the vehicle fluctuate and change slightly with the lateral movement of the car. The two variables converge to 0 again when the vehicle returns to the global planned path. It ensures a sufficient stability margin by minimizing the deviation of the replanned path.

Fig. 11(c) shows the maximum change rate of the longitudinal speed is less than 1% due to the influence of lateral motion. The variation ratio of the speed is within a reasonable range. Fig. 11(d) shows the results of the steering angle of the front wheel and the rear wheel of the autonomous vehicle.

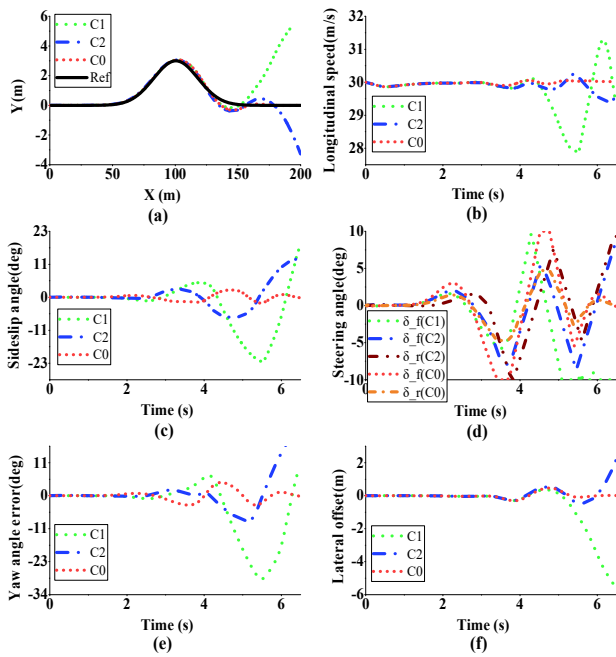


Fig. 12. HIL simulation results at 30m/s. (a) Tracking path. (b) Longitudinal speed. (c) sideslip angle of C.G. (d) Wheel steering angle. (e) Heading angle error. (f) Lateral offset error.

TABLE II
PATH FOLLOWING AND HANDLING STABILITY INDEX: RMS AND MAXIMUM ERROR FOR HIGH-SPEED CASE UNDER THREE CONTROL STRATEGIES

Control strategies	$\Delta\phi_{rms}$ (deg)	$ \Delta\phi_{max} $ (deg)	ΔY_{rms} (m)	$ \Delta Y_{max} $ (m)
C1	23.4300	29.22	0.1874	5.44
C2	19.8259	25.21	0.1368	3.36
C0	8.9560	4.58	0.0314	0.47

The autonomous car has accomplished the mission of collision avoidance and returned to the original planned path by controlling the steering wheels. The maximum steering of the front wheel is 5deg. It is worth mentioning that the steering direction of the front wheel is opposite to that of the rear wheel.

Case 2: High-speed 30m/s

To demonstrate the ability in terms of path following and handling stability of the vehicle at high speed, the test scenarios are executed that the planned path is set to double lane changing under the dry asphalt road with a constant speed 30m/s. The same test condition is used to test two other control methods, C1 and C2. Fig. 12 shows the HIL simulation results of three different control methods, C0, C1, and C2.

Fig. 12(a) shows the path tracking results of the autonomous vehicle under three different control strategies. The tracking path under the proposed control strategy can track the planned path, and no serious deviation and instability has occurred. Its maximum lateral tracking error is less than 0.5m. However, the tracking path of the autonomous vehicle under C1 or C2 severely deviates from the planned path, and the stability of the car cannot be guaranteed after 150m. The following trajectory of C1 has deviated from the planned path to 5.4m at the end of

path tracking HIL simulation, while the lateral tracking error under C2 has reached 3.3m.

Fig. 12(b) shows that the longitudinal speed varies with the path tracking under the three control strategies. Further comparisons show that the longitudinal speed of the proposed control strategy is most stable, and its change is less than 0.2m/s, compared with 2.3m/s and 0.7m/s of C1 and C2, respectively.

Fig. 12(c) shows the variation of sideslip angle of C.G changing with time, which fluctuates above and below zero under the proposed strategy when the autonomous vehicle tracks the path. The sideslip angle of C.G under C1 and C2 starts to oscillate with a larger amplitude after 4s in the high-speed condition, whose values almost reach 23deg and 13deg, respectively. The sideslip angle of C.G under C0 can eventually reach zero.

Fig 12(d) shows the simulation results of the front wheel steering angle and the rear wheel steering angle (if any). The front-wheel steering angle of C1 reaches its maximum limit that the large lateral offset occurred from 150m. The autonomous vehicle expects to ensure proper path tracking by adjusting the front wheel angle from 5s to 6.5s. However, due to the excessive sideslip angle, the car loses its stability, resulting in poor path tracking. For C2, the rear-wheel steering angle was added to maintain the path tracking ability. The vehicle under C2 also loses its stability owing to handling stability not taken into account. The steering angle of the front wheel in the proposed C0 reaches its maximum 10deg at about 3.5s and 4.5s. The vehicle satisfies the handling stability of the car as much as possible at the cost of short-term path tracking deviation.

As shown in Fig. 12(e) and (f), the yaw angle error and the lateral offset error of C0 are relatively small. At the same time, C1 and C2, from simulation results, show the vehicle has severely deviated from the planned path. The maximum value and the root mean square (RMS) value of the yaw angle error and the lateral offset error of the autonomous vehicle under three different control strategies are shown in Table II. The RMS values of the yaw angle error under C1, C2, and C0, respectively, are about 9.79deg 5.72deg and 1.11deg. The RMS value of the lateral offset error under C0 is 0.03m, compared with 0.19m (C1) and 0.14m (C2).

Case 3: Low tire-road adhesion coefficient 0.3

The purpose of Case 3 is to verify the performance of the proposed control strategy under the low-adhesion factor. The planned path is set to be the same as Case 2 in the driverless scene. In Case 3, the longitudinal speed of the autonomous vehicle is set to the constant 20m/s with the tire-road adhesion coefficient 0.3. The comparison HIL simulation results of three different control methods, C0, C1, and C2 under $\mu = 0.3$, are shown in Fig. 13.

Fig. 13(a) shows the path tracking results of the autonomous vehicle in which the solid black line represents preset planned paths, and the green dash line, blue dot-dashed line, and red dotted line, respectively, are tracking trajectories of C1, C2, and C0. C1 and C2 paths cannot follow the preset planned path because they had deviated from the planned path of 2.9m and 1.3m, respectively. However, the C0 tracking path error is 0.15m, which shows better path tracking ability.

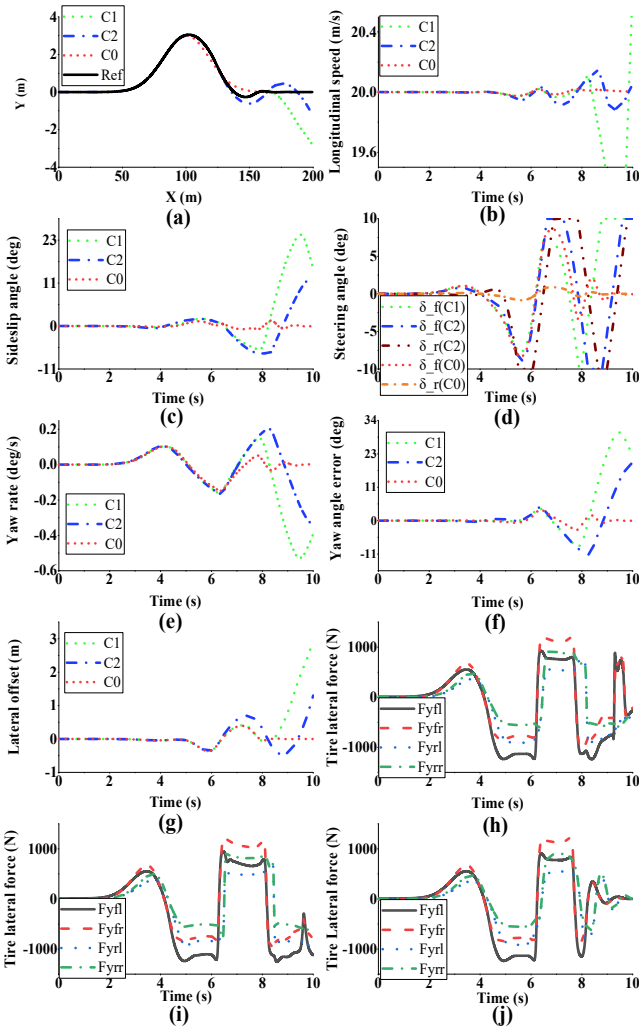


Fig. 13. HIL simulation results under $\mu = 0.3$. (a) Tracking path. (b) Longitudinal speed. (c) Sideslip angle. (d) Wheel steering angle. (e) Yaw rate. (f) Yaw angle error. (g) Lateral offset error. (h) Lateral force of each tire (C1). (i) Lateral force of each tire (C2). (j) Lateral force of each tire (C0).

TABLE III

PATH FOLLOWING AND HANDLING STABILITY INDEX: RMS AND MAXIMUM ERROR FOR LOW ADHESION CASE UNDER THREE CONTROL STRATEGIES

Control strategies	$\Delta\phi_{rms}$ (deg)	$ \Delta\phi_{max} $ (deg)	ΔY_{rms} (m)	$ \Delta Y_{max} $ (m)
C1	9.7925	30.36	0.6566	2.94
C2	5.7243	20.05	0.3158	1.30
C0	1.1059	3.43	0.1422	0.40

Fig. 13(b) show the longitudinal speed of the autonomous vehicle of C1, C2, and C0. The longitudinal speed of the proposed control strategy is most stable, compared with C1 and C2. The sideslip angle of C.G (C1) exceeds 23deg at 9.2s, while the sideslip angle of C.G (C2) reached 11.5deg at 10s, as shown in Fig. 13(c). The sideslip angle of C.G (C0) fluctuates slightly at a stable value, which shows the excellent performance of path tracking.

Fig. 13(d) shows the variation of the steering angle of the wheel under the three different control strategies. The vehicle under C1 is controlled to track the preset planned path by

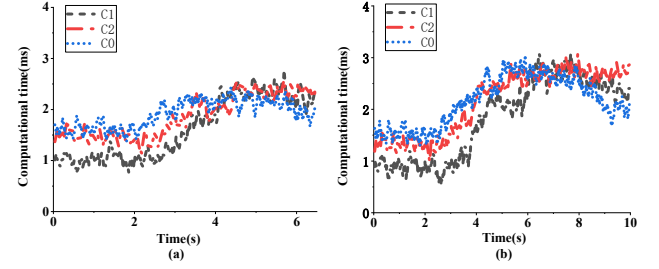


Fig. 14. Computation time in the HIL platform. (a) Computing time of Case 2 (high speed). (b) Computing time of Case 3 (Low tire-road adhesion coefficient).

quickly adjusting the front wheel angle when the car returns to the second straight line in the 7th second. However, the maximum value of the steering angle is reached in the 8th second, and the front wheel is turned to the other direction, which makes the adjustment meaningless for the vehicle's path tracking. In order to ensure the path tracking ability, the steering control of the rear wheel is introduced in C2. Its changing trend is similar to that of the front wheel. The steering wheel turns direction twice and reaches the limit value in the steering process. The rear wheel angle (C0) has a positive proportional relationship with the front wheel angle. It can ensure the path tracking of the vehicle and prevent the vehicle from losing stability due to the excessive sideslip angle by using adaptive wheel steering.

Fig. 13(e) shows the yaw rates of the vehicle of C1 and C2 are unable to maintain stability by controlling the steering angle. The yaw rate of C0 can keep to zero after the path tracking. Fig. 13(f) and (g) show the yaw angle error and lateral offset error, respectively. The yaw angle error (C0) fluctuates slightly, and the maximum lateral offset is 0.4m, which shows that the tracking trajectory can track the planned path well. The yaw angle error and the lateral offset error (C1 and C2) cannot stabilize to zero after the vehicle passes through double lane shifting. The maximum and RMS errors of the yaw rate and lateral offset are shown in Table III.

Fig. 13(h)-(j) show the lateral force of each tire under C1, C2, and C0, respectively. The changing trend and numerical value of the lateral force of under C1 and C2 are the same, respectively, which lead to small yaw moment by lateral tire force acting on the vehicle. The proposed control method C0 can coordinate the tire force of each tire needed for the vehicle motion to achieve path tracking as well as handling stability based on zero sideslip angle control.

The compared computation time for the proposed control algorithm (C0) and the other two algorithms (C1 and C2) in the HIL platform under Case 2 and Case 3 conditions is shown in Fig. 14. It can be seen from Fig.14(a) that their computation time is less than 2.5ms and never greater than 3ms through the simulation, which can satisfy the requirements of the real-time of the vehicle control with respect to the sampling time. The computational cost of C2 and C0 is considerable higher than that of C1 from 0 to 2s. During 2s to 4s, in order to track the desired trajectory in the lateral position, their computation time increases accordingly, and the increase of computing load of C1 is most obvious. During 4s to 6s in the HIL simulation, the computation time of C1 and C2 exceeds that of C0 slightly where it mainly because the tire steering angle is saturated due to the solution of the two control algorithms.

A similar phenomenon of the computation time can be observed in Fig. 14(b), where the difference is their computation time (Case 3) is slightly longer than the computation time (Case 2) in the second half of the HIL simulation time because the saturation of tire lateral force affects the calculation speed of the optimization function algorithm in the vehicle controller on the low adhesion road surface. Guo *et al.*, indicated the path-tracking error of the autonomous vehicle increases with a decrease in the sampling time [24]. However, the shorter sampling interval and more sophisticated algorithms mean the higher computation burden [45]. The proposed algorithm can achieve a better trade-off among handling stability, tracking performance, and the computation burden.

V. CONCLUSION

A novel control strategy with collision avoidance is presented for the highly automated vehicle, which aims to enhance the performance of path tracking and handling stability as well as obstacle avoiding capability under critical maneuvers.

The real-time and effectiveness of the proposed control strategy are verified in the HIL real-time simulation platform. The vehicle replans a new collision-free path when it needs to avoid an obstacle vehicle with a width of 2m in *Case 1*. The new replanned path ensures the safety margin of 0.5m when the car tracks the replanned trajectory under the road adhesion 0.3. Extensive experiments are carried out at high-speed 30m/s (*Case 2*) and under low tire-road adhesion coefficient 0.3 (*Case 3*). The maximum lateral tracking errors under two conditions are 0.47m/0.4m, compared with 5.44m/2.94m and 3.36m/1.3m of two other control methods. The proposed method exhibits excellent path tracking ability and handling stability, compared with two other methods. The computation time during the simulation remained below the sampling interval demonstrates it can satisfy the real-time requirements for the HIL simulation system.

In future work, the tire-road adhesion coefficient will be estimated in real-time to increase the robustness of the proposed control strategy. And the real-time and effectiveness of the proposed control algorithm need to be further evaluated and demonstrated on the experimental autonomous vehicle.

REFERENCES

- [1] A. Taeiagh and H. S. M. Iim, "Governing autonomous vehicles: emerging responses for safety, liability, privacy, cybersecurity, and industry risks," *Transport Rev.*, vol. 39, no. 1, pp. 103–128, JAN. 2019.
- [2] F. E. *et al.*, "Autonomous driving, the built environment and policy implications," *Transp. Res. Pt. A-Policy Pract.*, vol. 122, pp. 162–172, APR. 2019.
- [3] Hu. *et al.*, "Lane keeping of autonomous vehicles based on differential steering with adaptive multivariable super-twisting control," *Mech. Syst. Signal Process.*, vol. 125, pp. 330–346, JUN. 2019.
- [4] M. D. *et al.*, "Development and transport implications of automated vehicles in the Netherlands: scenarios for 2030 and 2050," *Eur. J. Transport. Infrastruct. Res.*, vol. 17, no. 1, pp. 65–85, JAN. 2017.
- [5] G. E. *et al.*, "Planning for Cars That Drive Themselves: Metropolitan Planning Organizations, Regional Transportation Plans, and Autonomous Vehicles," *J. Plan. Educ. Res.*, vol. 36, no. 2, pp. 210–224, JUN. 2016.
- [6] F. D. J. *et al.*, "Preparing a nation for autonomous vehicles: opportunities, barriers and policy recommendations," *Transp. Res. Pt. A-Policy Pract.*, vol. 77, pp. 167–181, JUL. 2015.
- [7] C. K. Dey *et al.*, "A review of communication, driver characteristics, and controls aspects of Cooperative Adaptive Cruise Control (CACC)," *IEEE Trans. Intell. Transp. Syst.*, vol. 17, no. 2, pp. 491–509, FEB. 2016.
- [8] S. Cheng *et al.*, "Multiple-Objective Adaptive Cruise Control System Integrated With DYC," *IEEE Trans. Veh. Technol.*, vol. 68, no. 5, pp. 4550–4559, MAY. 2019.
- [9] Z. Shuai, H. Zhang, J. Wang, J. Li and M. Ouyang, "Combined AFS and DYC control of four-wheel-independent-drive electric vehicles over CAN network with time-varying delays," *IEEE Trans. Veh. Technol.*, vol. 63, no. 22, pp. 591–602, MAY. 2014.
- [10] S. Y. Wei *et al.*, "An integrated longitudinal and lateral vehicle following control system with radar and vehicle-to-vehicle communication," *IEEE Trans. Veh. Technol.*, vol. 68, no. 2, pp. 1116–1127, FEB. 2019.
- [11] J. C. Li *et al.*, "RSE-Assisted Lane-Level Positioning Method for a Connected Vehicle Environment," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 7, pp. 2644–2656, JUL. 2019.
- [12] D. Kim, "Lane-Level Localization Using an AVM Camera for an Automated Driving Vehicle in Urban Environments," *IEEE/ASME Trans. Mechatro.*, vol. 22, no. 1, pp. 280–290, FEB. 2017.
- [13] J. Q. Nie *et al.*, "Decentralized cooperative Lane-Changing decision-making for connected autonomous vehicles," *IEEE Access*, vol. 4, pp. 9413–9420, DEC. 2016.
- [14] G. D. *et al.*, "A Review of Motion Planning Techniques for Automated Vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 17, no. 4, pp. 1135–1145, APR. 2016.
- [15] S. macheng, S. Jing, P. Huei and Z. Ding, "Improving localization accuracy in connected vehicle networks using rao-blackwellized particle filters: theory, simulations, and experiments," *IEEE Trans. Veh. Technol.*, vol. 20, no. 6, pp. 2255–2266, JUN. 2019.
- [16] J. Ji *et al.*, "Path planning and tracking for vehicle collision avoidance based on model predictive control with multiconstraints," *IEEE Trans. Veh. Technol.*, vol. 66, no. 2, pp. 952–964, FEB. 2017.
- [17] J. H. Chuan *et al.*, "Robust H-infinity output-feedback control for path following of autonomous ground vehicles," *Mech. Syst. Signal Process.*, vol. 70–71, pp. 414–427, MAY. 2016.
- [18] M. Klomp *et al.*, "Trends in vehicle motion control for automated driving on public roads," *Veh. Syst. Dyn.*, vol. 57, pp. 1028–1061, JUL. 2019.
- [19] X. H. Li *et al.*, "Development of a new integrated local trajectory planning and tracking control framework for autonomous ground vehicles," *Mech. Syst. Signal Process.*, vol. 87, no. SI, pp. 118–137, MAR. 2017.
- [20] C. JB *et al.*, "Effectiveness of forward collision warning and autonomous emergency braking systems in reducing front-to-rear crash rates," *Accident Analysis and Prevention*, vol. 99, pp. 142–152, FEB. 2017.
- [21] J. Funke *et al.*, "Collision Avoidance and Stabilization for Autonomous Vehicles in Emergency Scenarios," *IEEE Trans. Contr. Syst. Tech.*, vol. 25, no. 4, pp. 1204–1216, JUL. 2017.
- [22] X. K. He *et al.*, "Emergency steering control of autonomous vehicle for collision avoidance and stabilisation," *Veh. Syst. Dyn.*, vol. 57, no. 8, pp. 1163–1187, AUG. 2019.
- [23] S. Cheng *et al.*, "Longitudinal Collision Avoidance and Latera Stability Adaptive Control System Based on MPC of Autonomous Vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 15, no. 5, pp. 1–10, MAY. 2018.
- [24] H. Y. Guo *et al.*, "Model predictive path following control for autonomous cars considering a measurable disturbance: Implementation, testing, and verification," *Mech. Syst. Signal Process.*, vol. 118, pp. 41–60, MAR. 2019.
- [25] R. R. Wang *et al.*, "Composite nonlinear feedback control for path following of four-wheel independently actuated autonomous ground vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 17, no. 7, pp. 2063–2074, JUL. 2016.
- [26] N. R. Kapania, R. Nitin and C. J., "Design of a feedback-feedforward steering controller for accurate path tracking and stability at the limits of handling," *Veh. Syst. Dyn.*, vol. 53, no. 12, pp. 1687–1704, DEC. 2015.
- [27] H. L. Yuan *et al.*, "Unified decision-making and control for highway collision avoidance using active front steer and individual wheel torque control," *Veh. Syst. Dyn.*, vol. 57, no. SI, pp. 1188–1205, AUG. 2019.
- [28] K. Nam, S. Oh, H. Fujimoto, and Y. Hori, "Robust yaw stability control for electric vehicles based on active front steering control through a steer-bywire system," *Int. J. Automotive Technol.*, vol. 13, no. 7, pp. 1169–1176, Dec. 2012.
- [29] X. Z. Zhang, and X. L. Zhu, "Autonomous path tracking control of intelligent electric vehicles based on lane detection and optimal preview method," *Expert. Syst. Appl.*, vol. 121, pp. 38–48, MAY. 2019.

- [30] X. Q. Sun *et al.*, "Optimal control of intelligent vehicle longitudinal dynamics via hybrid model predictive control," *Robot. Auton. Syst.*, vol. 112, pp. 190–200, FEB. 2019.
- [31] P. Falcone *et al.*, "Predictive Active Steering Control for Autonomous Vehicle Systems," *IEEE Trans. Contr. Syst. Tech.*, vol. 15, no. 3, pp. 566–580, MAY. 2007.
- [32] K. Kritayakirana and J.C. Gerdes, "Autonomous vehicle control at the limits of handling," *Int. J. Veh. Auton. Syst.*, vol. 10, no. 4, pp. 271–296, DEC. 2015.
- [33] J. Ni and J.B. Hu., "Dynamics control of autonomous vehicle at driving limits and experiment on an autonomous formula racing car," *Mech. Syst. Signal Process.*, vol. 90, pp. 154–174, JUN. 2017.
- [34] C. Hu *et al.*, "Output Constraint Control on Path Following of Four-Wheel Independently Actuated Autonomous Ground Vehicles," *IEEE Trans. Veh. Technol.*, vol. 65, no. 6, pp. 4033–4043, JUN. 2016.
- [35] T. Chen *et al.*, "Simultaneous path following and lateral stability control of 4WD-4WS autonomous electric vehicles with actuator saturation," *Adv. Eng. Softw.*, vol. 128, pp. 46–54, FEB. 2019.
- [36] Shengbo E. L. *et al.*, "Predictive lateral control to stabilise highly automated vehicles at tire-road friction limits," *Veh. Syst. Dyn.*, vol. 58, pp. 768–786, Jan. 2020.
- [37] Y.J. Hu *et al.* "A Motion Planning and Tracking Framework for Autonomous Vehicles Based on Artificial Potential Field Elaborated Resistance Network Approach," *IEEE Trans. Ind. Electron.*, vol. 67, no. 2, pp. 1376–1386, FEB. 2020.
- [38] J. J. Hu *et al.* "Lane Detection and Trajectory Tracking Control of Autonomous Vehicle Based on Model Predictive Control," *Int. J. Automot. Technol.*, vol. 21, no. 2, pp. 285–295, APR. 2020.
- [39] C. Hu *et al.* "MME-EKF-Based Path-Tracking Control of Autonomous Vehicles Considering Input Saturation," *IEEE Trans. Veh. Technol.*, vol. 68, no. 6, pp. 5246–5259, JUN. 2019.
- [40] H.B. Ren *et al.*, "Effective assessment of tyre–road friction coefficient using a hybrid estimator," *Veh. Syst. Dyn.*, vol. 52, pp. 1047–1065, APR. 2014.
- [41] Y. Chen *et al.*, "Optimized handling stability control strategy for a four in-wheel motor independent-drive electric vehicle," *IEEE Access.*, vol. 7, pp. 17017–17032, JUN. 2019.
- [42] M. Doumiani, "Observers for vehicle tyre/road forces estimation: experimental validation," *Veh. Syst. Dyn.*, vol. 48, no. 11, pp. 1345–1378, OCT. 2010.
- [43] H. Zhang *et al.*, "Robust gain-scheduling energy-to-peak control of vehicle lateral dynamics stabilisation," *Veh. Syst. Dyn.*, vol. 52, no. 3, pp. 309–340, MAR. 2014.
- [44] O. Gietelink *et al.*, "Development of advanced driver assistance systems with vehicle hardware-in-the-loop simulations," *Veh. Syst. Dyn.*, vol. 44, no. 7, pp. 569–590, FEB. 2007.
- [45] E. Siampis *et al.*, "A Real-Time Nonlinear Model Predictive Control Strategy for Stabilization of an Electric Vehicle at the Limits of Handling," *IEEE Trans. Contr. Syst. Tech.*, vol. 26, no. 6, pp. 1982–1994, JUL. 2018.



Yong Chen received his M.E. degree in School of Mechanical Engineering, Beijing Institute of Technology, Beijing, China in 2018. He is currently working toward the Ph.D. degree in the School of Mechanical Engineering, Beijing Institute of Technology. His research interests include electric vehicles, vehicle dynamic and control and autonomous vehicles as well as relevant technologies of intelligent vehicles.



design and control technology.

Sizhong Chen received his M.S. degree in College of Automotive Engineering, Jilin University, Jilin, China in 1987. Now, he is currently a professor of Automotive Research Institute in School of Mechanical engineering, Beijing Institute of Technology.

He serves as the secretary-general of the SUV Technology Committee of Society of Automotive Engineering of China. His current research interests include suspension dynamics and steering, vehicle



Hongbin Ren received Ph. D degree in Mechanical Engineering from Beijing Institute of Technology. He studied at the University of Michigan in Dearborn, Dearborn, USA, from 2012.10 to 2014.04 as a visiting scholar.

His current research interests include vehicle dynamics, state observer theory, optimization, and optimal control.



Zepeng Gao received his M.S. degree in School of Mechanical engineering, Beijing Institute of Technology, Beijing, China in 2017. Now he is a Ph.D. candidate at Beijing Institute of Technology.

His main research area is vehicle dynamics theory and control, which includes air suspension height adjustment and application of vehicle system vibration energy recovery technology as well as intelligent control algorithm in vehicle dynamics.



Zheng Liu received his M.S degree from University of Science and Technology, Beijing, in 2014. He is currently pursuing the Ph.D. degree with School of Mechanical Engineering, Beijing Institute of Technology.

His research interests include vehicle dynamic and control, semi-active/active suspension system, and relevant technologies of intelligent vehicles.